

QUALIFYING EXAM

An ontology-based approach to accelerate the reconstruction of genome-scale metabolic models

Nahim Alves de Souza

Supervisor
Profa. Dra. Renata Wassermann

Co-supervisor
Dra. Mariana Galvão Ferrarini

August, 2025

Contents

Abstract . . . 2
Introduction . . . 3

1 GEM Reconstruction **5**
1.1 Stages of GEM Reconstruction . . . 6
1.2 Computational Resources for GEM Reconstruction . . . 9
1.2.1 Databases . . . 10
1.2.2 Tools . . . 11
1.3 Key Challenges in GEM Reconstruction . . . 14
1.3.1 Data Representation Issues . . . 14
1.3.2 Experimental and Literature-based Model Validation . . . 18
1.3.3 Research Fragmentation and Collaboration Barriers . . . 19

2 Ontologies and Semantic Interoperability **20**
2.1 Ontologies . . . 20
2.1.1 Ontology Development Methodologies . . . 21
2.1.2 Ontologies for Biology and Chemistry . . . 22
2.2 Semantic Interoperability . . . 23
2.2.1 Enabling Interoperability with Ontologies . . . 23

3 Research Proposal: An Ontology for GEMs **25**
3.1 Motivation and Expected Contributions . . . 25
3.2 Ontology Development . . . 27
3.2.1 Competency Questions . . . 27
3.2.2 Conceptualization . . . 28
3.2.3 Formalization and Semantic Enrichment . . . 31
3.3 Ontology-Based Application . . . 32
3.4 Evaluation of Ontology-Based Reconstruction . . . 32
3.5 Challenges and Limitations . . . 33

4 Project Timeline and Work Plan **35**
4.1 Activities Description . . . 35

Abstract

Genome-scale metabolic models (GEMs) provide comprehensive views of metabolic processes and enable simulations of metabolic activity. However, their reconstruction is a complex and labor-intensive task, heavily reliant on expert manual curation. This dependence not only constitutes a major bottleneck but also increases the risk of inconsistencies. The process involves integrating data from diverse sources, such as biochemical databases and scientific literature, which often contain inconsistencies due to the lack of standardized representations for metabolites and reactions. While existing solutions – including integrated datasets, controlled vocabularies, taxonomies, and ontologies – provide partial support, they remain insufficient for fully addressing the challenges of data integration and system interoperability. Despite this, domain experts are able to resolve most of these inconsistencies manually using their background knowledge, which suggests current limitations probably stem from the absence of explicit semantic information in the data. Numerous studies reinforce this hypothesis by demonstrating that semantically enriched models are effective in facilitating the integration of heterogeneous data sources and systems. Therefore, we believe that a semantically enriched model for representing GEMs, such as a domain-specific ontology, could address these interoperability challenges. To test this hypothesis, this work proposes developing such an ontology and integrating it into the GEM reconstruction workflow through a dedicated application. This application will leverage the ontology to integrate data from multiple biochemical datasets, producing enhanced models that will be evaluated according to expert-defined criteria. Beyond improving data integration, this semantically enriched approach is expected to enable automated reasoning, enhance knowledge sharing, and facilitate model reuse, ultimately reducing manual curation efforts and streamlining the entire GEM reconstruction process.

Introduction

Metabolism is the set of chemical processes occurring continuously within living organisms to sustain life [1]. Such biological processes are studied by Systems Biology experts from a systemic perspective, using computational models and simulations to provide a in-depth understanding of the metabolism of an organism. This comprehension of metabolism can have applications in several fields [2–4], such as the identification of biomarkers for diseases, the detection of essential metabolites that could serve as potential drug targets, and the analysis of changes in metabolic reactions in response to certain substances or environmental changes.

Definição de GEM

Nevertheless, mapping the complete metabolism of an organism is a highly com… vast number of compounds and reactions involved. High-throughput experiments (generating large volumes of data) and sophisticated mathematical analyses are required to identify the compounds consumed and produced by the organism. The process of building a computational model of an organism's metabolism is known as **reconstruction**, while the resulting model is often referred to as a **genome-scale metabolic model (GEM)**, since it is based on organism's genome – other names and acronyms found in the literature (e.g, genome-scale metabolic networks, GSMN, GSMM, and GSM) can usually be treated as synonyms.

Over the years, a large number computational resources have been developed to support the reconstruction of metabolic models. These include databases that store chemical and biological data, as well as tools for simulating metabolic fluxes and performing mathematical analyses [5]. However, significant limitations remain, particularly in the unification of data from multiple sources [6–10], which keeps the process heavily reliant on manual efforts from domain experts. One of the major limitations is the existence of different naming conventions and identifier formats for biochemical data, which depend on the database [6, 11] or the tool used to build the metabolic model [12].

Referências sobre integração de dados de fontes múltiplas com o uso de ontologias

…ve achieved good results, across various domains, in integrating data from mul… …he use of ontologies [13–17]. A unified representation for metabolic models …ology experts combine and compare models, and reconcile biochemical data … Experts currently rely on their implicit domain knowledge to manually resolve …which suggests that a semantically enriched representation containing this type information could be used to address these issues. In additi…

Possibilidade de gerar perguntas que desafiem o LLM a identificar inconsistências nos dados

a more accurate representation of metabolic models, accel… foster new scientific discoveries.

Despite the existence of several ontologies for biochemica… …, GEMs or are actively employed in GEM reconstruction. Current ontologies usually focus on representing hierarchical structures rather than capturing biological functions and interactions. For example, Gene Ontology (GO) [18] describes gene functions and products, ChEBI [19] represents chemical entities of biological interest and their relationships, and SBO [20] defines various Systems Biology

terms. Nevertheless, the complex relationships between reactions, metabolites, and genes – essential for integrating data during GEM reconstruction – are not fully captured by any of them.

Based on these premises, this work proposes the development of a new ontology for representing GEMs with two main objectives: (1) to enable the integration and reconciliation of models across different datasets, and (2) to facilitate the evaluation of GEMs through the use of logical inferences to identify (and potentially repair) inconsistencies in the network.

As a result of this work, at least two artifacts must be produced to achieve the established goals: (1) an ontology for representing GEMs, which includes genes, metabolites and reactions, thereby providing the necessary information to support the comparison and integration of models from different datasets; and (2) a user-friendly tool serving as a proof of concept (PoC) to ensure the ontology integration into the GEM reconstruction workflow, validate the ontology's applicability and evaluate its effectiveness in simplifying the GEM reconstruction process.

Demonstração de como descrever o que significa o sucesso do trabalho

The remainder of this document presents the GEM reconst es, explains how a specialized ontology can help address these c tu-alization of the domain, and discusses the key benefits and lim

Chapter 1

GEM Reconstruction

The set of chemical processes that occur within the cells of living organisms for supporting energy production, storage, and consumption is called **metabolism**. These processes are responsible for the synthesis of small molecules, such as amino acids, nucleotides, and simple sugars, which serve as building blocks for larger molecules essential for growth, repair, and other vital cellular functions [1].

The substances produced or consumed by the metabolic processes are known as **metabolites**, which can be classified as primary or secondary according to their function in the organism. **Primary metabolites** are directly involved in organism growth, development and reproduction, and they are present in most organisms. On the other hand, **secondary metabolites** are usually not essential for life, but have important ecological functions, such as protecting plants against insects, fungi, and herbivores, or helping them to attract pollinators [21].

Metabolic reactions are biochemical processes involved in the transformation of molecules, including binding, dissociation, modification, and transport. They are categorized into **catabolic reactions**, which break down macromolecules to release energy, and **anabolic reactions**, which consume energy to synthesize new macromolecules. These reactions typically require **enzymes**, biological catalysts (usually proteins) that initiate or accelerate metabolic processes, but are not consumed by them.

A set of interconnected metabolic reactions involved in a related process is commonly referred to as a **metabolic pathway** [1]. In such pathways, the product of one reaction often serves as a substrate for the next. However, some products may not be utilized by any subsequent reactions in the organism and are therefore considered **dead-end metabolites**. A collection of metabolic pathways within a cell or organism forms a **metabolic network** and its visual representation is typically known as a **metabolic map**. The complete set of metabolites within a biological sample (such as a cell, tissue, or organism) is called **metabolome**.

In the study of metabolism, computational metabolic models (also known as *in silico* models) can be used to represent metabolic reactions in a simplified manner, by focusing on their key components. These models facilitate the aggregation and organization of knowledge, promote a deeper understanding of living organisms and serve as valuable tools for biological simulations, through which important components and interactions can be identified [22].

Depending on the research objective, different types of computational metabolic models can be employed. For example, **Genome-Scale Metabolic Models (GEMs)** – also known as Genome-scale Metabolic Networks (GSMNs) or Genome-scale Metabolic Graphs (GSMGs) [23] – are based on the organism's genome and are specifically designed to provide a holistic representation of the its metabolism. These models can be used to simulate growth, predict metabolic behavior across various environments

[24], and perform tasks such as analyzing energy production (ATP generation), optimizing metabolite production, conducting gene essentiality tests, and identifying potential drug targets [25].

Although GEMs have proven to be a powerful tool for enhancing our understanding of complex biological processes, they still have limitations [25]. For example, GEMs cannot fully capture all aspects of complex organisms, such as eukaryotes, due to the absence of complete genome annotations and the intricacies of their intracellular organization. In addition, GEM constraint-based modeling and flux simulations typically assume steady-state conditions, whereas real biological environments are dynamic and subject to change over time. Furthermore, many of these methods rely on linear programming and optimization techniques, which may not always produce deterministic solutions and often require additional evaluation using sophisticated (and computationally expensive) methods to identify optimal outcomes.

1.1 Stages of GEM Reconstruction

The process of building a genome-scale metabolic model is known as **reconstruction**. It involves a sequence of carefully executed steps to ensure the quality and accuracy of the model. Thiele and Palsson [26] proposed a detailed five-stage protocol for constructing high-quality GEMs, which is summarized in Figure 1.1 and described below:

1. **Draft Reconstruction** – Once an organism of interest is selected, a draft model is constructed based on its annotated genome and biochemical databases, in a process known as ***de novo*** reconstruction. Tools such as CarveMe [27] and ModelSEED [28] can generate this initial model automatically by identifying candidate metabolic functions through various approaches, such as matching Enzyme Commission (EC) numbers[1] or through homology and orthology analysis[2]. Afterward, metabolic reactions for each function are mapped by linking gene information to biochemical databases, such as KEGG [32], BRENDA [33], or BiGG [34].

   Alternatively, the draft model can be constructed through a **template-based** approach [35], which consists in applying homology-based methods to the template to infer the reactions that should be included in the model [31]. The template may be a highly curated GEM from a closely related species or a partially complete GEM that requires further refinement. In some cases, multiple GEMs can be merged to incorporate data from diverse sources, for example, models developed for different strains of the same organism, closely related species, or by different research groups. Tools like MetaNetX [36] and *mergem* [9] facilitate this type of integration.

   The draft model is encoded into a single file that associates genes with enzymes, reactions, metabolites and additional annotations, representing a metabolic network. This file is typically a spreadsheet or an **SBML (Systems Biology Markup Language)** file – an XML-based standard compatible with most tools used in the GEM reconstruction process [37]. At this point, the model will likely contain errors due to missing, incorrect, or incomplete data, which makes thorough manual curation essential to ensure its accuracy and reliability.

2. **Manual Refinement** – In this stage, all entries in the draft model are carefully inspected to ensure they were correctly included in the draft, and that the relationships among them are ac-

[1] EC numbers are a classification scheme to categorize enzymes based on the reactions they catalyze [29].

[2] These analyses aim to determine whether two organisms share a common ancestor, which may indicate that they have similar metabolic functions and reactions [30, 31].

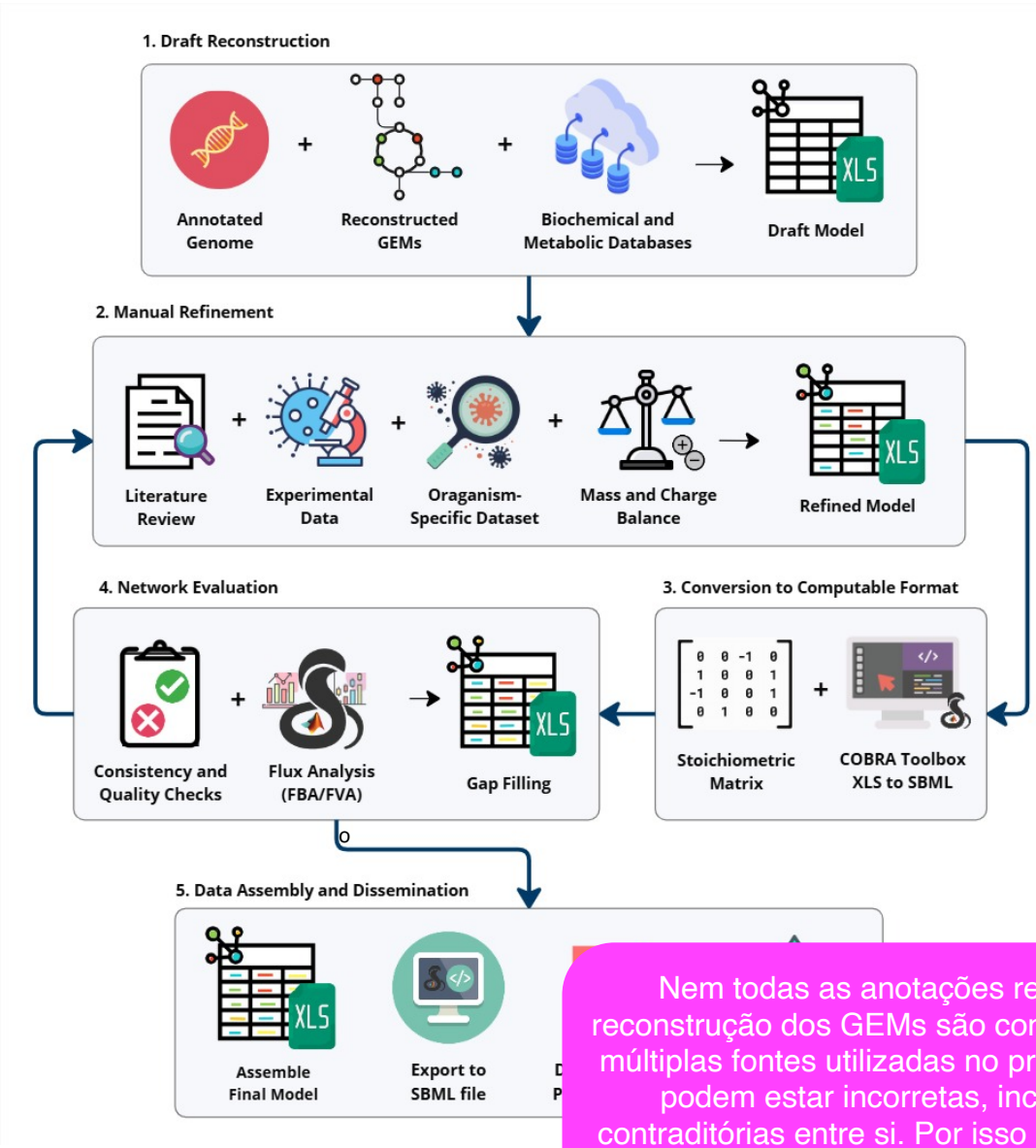


Figure 1.1: GEM Reconstruction stages, bas

Nem todas as anotações resultantes da reconstrução dos GEMs são confiáveis devido às múltiplas fontes utilizadas no processo, as quais podem estar incorretas, incompletas ou contraditórias entre si. Por isso é importante que se coletem evidências na literatura científica e/ou experimentos para suportar as anotações e garantir a sua qualidade.

curate. This means incorrect entries can be removed, and missing connections should be added to improve model accuracy. Given that not all annotations are reliable, supporting evidence from scientific literature or biological experiments must be gathered to ensure the quality of the reconstructed model.

Thiele and Palsson's protocol [26] describes several aspects that should be analyzed in this stage, such as verifying substrate and cofactor usage, determining charged formulas for metabolites, calculating reaction stoichiometry, determining reaction directionality, establishing gene and protein localization, validating gene-protein-reaction (GPR) associations, adding metabolite identifiers from public databases (e.g, KEGG [32], ChEBI [38], MetaNetX [36]) and database-independent representations (e.g., SMILES, InChI), assigning confidence scores[3], determining biomass composition, and incorporating missing reactions (e.g., spontaneous, intracellular transport, ATP maintenance, demand and sink reactions).

[3]In their work [26], Thiele and Palsson proposed a confidence score system, where scores range from 0 to 4 (low to high confidence, respectively), according to the amount of evidences that support each information.

3. **Conversion to Computable Format** – While the spreadsheet format facilitates manual curation by allowing data to be easily visualized and edited, other formats, such as SBML, are more suitable for conducting computational analyses [5]. For example, for performing mathematical analyses, such as **FBA (Flux Balance Analysis)**, a **stoichiometric matrix (S Matrix)** is usually the best structure to represent the metabolic network. The conversion is a straightforward process, as tools like MATLAB COBRA Toolbox[4] allow the direct import of data from spreadsheets or SBML files and automate the conversion into this format.

   FBA is a widely adopted method for studying GEMs [22, 41]. It uses linear programming methods to calculate the flux distribution that maximizes or minimizes the production of specific metabolites within a constrained environment. For example, FBA can be used to analyze the relationship between a given metabolite and energy production by defining the reactions that generate ATP as the model's objective function (a binary vector specifying the reactions of interest), and applying uptake rate constraints to simulate physiological conditions. The method then computes a flux distribution vector that optimizes the objective function, predicts the amount of ATP produced under these conditions and indicates the contribution of each metabolite to the process [41].

   The S Matrix is a representation in which each entry indicates how metabolites (rows) are consumed or produced by reactions (columns) under steady-state conditions. Beyond its mathematical formulation, the S matrix also captures essential characteristics of the metabolic network. Each column corresponds to a stoichiometrically balanced reaction, imposing constraints on metabolite flow, while each row reflects the connectivity between reactions, indicating which reactions produce or consume a given metabolite. Due to these properties, the S matrix plays a fundamental role in metabolic models and enables the exploration of various aspects of the metabolic network [42].

4. **Network Evaluation** – Since numerous network gaps and missing functions are expected to be found during the model curation, a suite of tests is necessary to evaluate and debug the reconstructed metabolic network. The testing process addresses three main objectives: **validation**, ensuring the model achieves its intended purpose; **verification**, confirming that the model accurately represents the biological system it is designed to simulate; and **evaluation**, measuring the effectiveness and reliability of the model in performing its designated tasks.

   Most tests focus on identifying and addressing missing reactions or metabolites, while assessing the model's ability to synthesize specific products. This process, known as **gap analysis**, provides valuable insights into the metabolic model and helps uncover inconsistencies such as stoichiometric balanced cycles, dead-end metabolites (metabolites that are not consumed by any reaction) and blocked reactions (reactions unable to carry any flux). The issues identified during this process are addressed through a procedure known as **gap filling**, which aims to add missing reactions to the model in order to minimize the number of gaps.

   The primary goal of gap filling is to ensure that the model is functional; therefore, the number of reactions added to address gaps should be minimum. Attention is necessary to avoid introducing reactions or functions that do not exist in the actual organism. Counterintuitively, adding new reactions to the network may create additional gaps. As such, the inclusion of candidate

[4] MATLAB is a proprietary software, but there are free open-source alternatives to perform constrained-based reconstruction and analysis, e.g., COBRApy [39, 40].

reactions and genes should be limited to those related to essential functions and supported by experimental evidence. If the model produces unexpected metabolites due to incorrectly added reactions, those reactions should be removed. However, since removing a single reaction may not always eliminate the unintended synthesis of such metabolites, regulatory information and other possible causes must also be considered.

Tests must also verify the model's ability to produce biomass components and specific metabolites, as incorrectly added reactions can affect the predicted growth rate. If growth is lower than expected, it may indicate that key metabolite precursors are not being sufficiently synthesized. Conversely, if growth is higher than expected, this could suggest the unintended inclusion of reactions or metabolic cycles. In both cases, further investigation is necessary to identify the underlying causes and correct the reconstructed model. Biochemical textbooks and databases are valuable resources for determining known production capabilities (metabolites that should be produced by the organism) and known incapabilities (metabolites that should not be produced), and should therefore be consulted during this step.

Discrepancies between predicted and observed phenotypes or gene essentiality can highlight deficiencies in the model and guide the identification of missing or incorrect components in the metabolic network. Comparing **in silico** data with experimental data can also provide valuable insights into the model's ability to simulate organism behavior. Additionally, tools like MEMOTE [43] can be used to perform consistency and quality checks, offering a complementary suite of tests to further evaluate model performance.

5. **Data Assembly and Dissemination** – Ultimately, all reconstruction data must be compiled, and a biological interpretation of the results should be provided. It is acceptable for the final model to include dead-end metabolites and blocked reactions, as long as they do not compromise the scope or the primary objectives of the reconstruction. Artifacts produced during the reconstruction should be made available at the end of the process[5], including: (1) a spreadsheet containing all information from the reconstructed model; (2) an SBML file representing the GEM; (3) a web-accessible version of the model, published in a biochemical database such as BiGG [34]; and (4) documentation describing the reconstruction process and results, along with a biological interpretation.

The proposed stages for GEM reconstruction are not intended to be rigidly followed, as the process can vary according to the focus of the metabolic reconstruction. For instance, stages 2 to 4 can be considered as a single one, since they are involved in model refinement. Additionally, the importance of each stage may vary according to the scenario. If no prior GEM exists or relevant data in databases are scarce, the literature review and biological experiments will become critical parts of the process. In contrast, if a large amount of data is available, the primary focus may be on consolidating information from various sources into a unified model before incorporating any new data.

1.2 Computational Resources for GEM Reconstruction

In summary, GEM reconstruction is a complex process that involves integrating data from diverse sources, conducting thorough literature reviews, performing expert manual curation, running mathematical simulations, and validating results through biological experiments. Various publicly available

[5] A good example of GEM reconstruction artifacts can be found in the supplementary materials of [44].

resources support this process and help to improve the quality of the final model. This section provide an overview of the databases and tools widely used for GEM reconstruction.

1.2.1 Databases

Due to the diversity of data, a single GEM reconstruction may require multiple databases to be consulted in order to gather all the necessary information. Table 1.1 provides a list of databases commonly used in GEM reconstruction, along with their respective scopes (i.e., the type of data it contains).

Table 1.1: Databases for GEM reconstruction. *Scope* refers to the type of data included in the database.

Database	**URL**	**Scope**
BiGG [34]	http://bigg.ucsd.edu	Metabolic models
KEGG [32]	https://www.kegg.jp	Genes, Reactions, Pathways, Metabolites
MetaCyc [45]	https://metacyc.org	Enzymes, Genes, Pathways, Metabolites
MetaNetX [36]	https://metanetx.org	Reactions, Pathways, Metabolites
BRENDA [33]	https://www.brenda-enzymes.org	Enzymes, Reactions, Pathways
Rhea DB [46]	https://www.rhea-db.org	Reactions
Reactome [47]	https://reactome.org	Pathways, Reactions
ChEBI [19]	https://www.ebi.ac.uk/chebi	Chemical compounds
MetaboLights [48]	http://metabolights.org	Metabolites
ModelSEED DB [28]	https://modelseed.org/biochem	Metabolites, Reactions
LIPID MAPS [49]	http://www.lipidmaps.org	Pathways, Reactions, Metabolites
SwissLipids [50]	http://www.swisslipids.org	Metabolites
HMDB [51]	https://hmdb.ca	Metabolites
UniProt [52]	https://www.uniprot.org	Genes, Proteins, Enzymes
PubChem [53]	https://pubchem.ncbi.nlm.nih.gov	Chemical compounds, Scientific literature
FORUM KG [7]	https://forum-webapp.semantic-metabolomics.fr	Chemical compounds, Scientific literature
WikiPathways [54]	https://www.wikipathways.org	Pathways
Bioregistry [8]	https://bioregistry.io	Unified URIs

While some databases, such as KEGG and MetaCyc, encompass multiple types of data (e.g., genes, pathways, diseases), others are specialized in a single category and use cross-references to external datasets to provide additional information (e.g., BRENDA, Rhea, Reactome). Some databases cover a wide variety of metabolites, while others focus on specific groups. For example, LIPID MAPS and SwissLipids specialize in lipids, HMDB focuses on the human metabolome, and UniProt on proteins. In contrast, ModelSEED, BiGG, KEGG, ChEBI and MetaboLights cover a diverse set of metabolites without a strict specialization.

Cross-references are widely used in biochemical databases. PubChem [53], an open chemistry database, aggregates data from hundreds of databases and organizes them into categories, such as substances, genes, pathways, patents, and literature. Its data is available in a machine-readable format called **RDF (Resource Description Framework)**, which enables the representation of semantic relationships between entities [55]. The FORUM KG database integrates information from PubChem and other sources into a knowledge graph in order to facilitate the discovery of associations between compounds and diseases, and to support biological data interpretations. Similarly, WikiPathways [54],

an open-source biological pathway database, uses cross-references to link pathway data with organisms, research communities, authors and annotations.

In addition to the use of cross-references, there are datasets specifically developed for data unification. For example, MetaNetX consolidates information from 11 different databases by reconciling identifiers, names, and chemical data into a unified namespace called MNXref [6]. By using this namespace, entities from different databases can be mapped to the same MetaNetX identifier, representing they are a single real-world entity, thereby reducing ambiguity across multiple data sources.

With a similar objective but a different approach, Bioregistry [8] – a component of the Biopragmatics Stack[6] – was developed to facilitate data integration by providing a service to standardize and map identifiers of biochemical entities. It consists of a dataset containing a registry and a meta-registry of compact URIs (CURIEs), along with a service that maps these CURIEs to content providers. By leveraging these features, cross-references can be standardized, which reduces data duplication and ambiguities.

Furthermore, some datasets not listed in the table, focus on specific groups of organisms aiming to streamline queries and simplify result visualization. For instance, the VEuPathDB Bioinformatics Resource Center[7] provides a collection of resources containing data related to a wide range of eukaryotic pathogens and their vectors, offering specialized views for specific subsets of its data. One of these datasets is the TriTrypDB [56], which is dedicated to data on *Trypanosoma* and *Leishmania* parasites.

1.2.2 Tools

Several tools have been developed to streamline the GEM reconstruction process (Table 1.2). These tools provide support for a variety of tasks, such as GEM analysis, reconstruction, validation, manipulation, visualization, and simulations. Similar functionalities can be found in different tools, which gives experts the flexibility to choose the tool that best align with their preferences or expertise. Moreover, additional resources, such as libSBML [57] and the SBML Test Suite[8] have been designed to facilitate reading, writing, manipulating, translating, and validating SBML files.

Although no single tool currently allows full automation of the GEM reconstruction process, existing tools can significantly reduce the workload and make the process less labor-intensive by streamlining specific reconstruction steps. A selection of commonly used tools for GEM reconstruction is listed and briefly described below, and summarized in Table 1.2.

- **The COBRA Toolbox** – A MATLAB plugin for Constraint-Based Reconstruction and Analysis (COBRA) used for modeling, analyzing and predicting metabolic phenotypes based on GEMs. It includes several methods to simulate and analyze the behavior of metabolic networks, e.g., FBA and FVA. The toolbox also incorporates features for model reconstruction (e.g., gap filling) and visualization. While the plugin itself is open-source, it requires MATLAB, a proprietary software that necessitates a paid license.

- **COBRApy** – A Python library that implements COBRA methods, developed by the openCOBRA group, the same team responsible for maintaining The COBRA Toolbox. Due to its free and open-source nature, and Python's widespread adoption, this library has been employed in the development of many applications, such as CNApy, Escher and MEMOTE[9].

[6] Available at: https://biopragmatics.github.io/
[7] Available at: https://veupathdb.org
[8] More information at: https://sbml.org/software/sbml-test-suite/.
[9] A list of packages that use COBRApy can be found at: https://opencobra.github.io/cobrapy/packages/.

- **CNApy** – An open-source Python-based tool with an integrated visual environment for metabolic modeling. It provides common COBRA methods, such as FBA and FVA, and enables model visualization and editing. Moreover, Escher maps are natively supported and can be created and edited inside the tool.

- **eQuilibrator** – An open-source web application designed for the thermodynamic analysis of biochemical reactions and pathways. It enables the estimation of thermodynamic parameters under various conditions, which can provide insights on how much energy a particular reaction requires and in which direction the reaction will flow.

- **The RAVEN Toolbox** – RAVEN is the acronym for "Reconstruction, Analysis and Visualization of Metabolic Networks", a MATLAB plugin that implements semi-automated draft model reconstruction by integrating data from KEGG database and existing template GEMs. The plugin provides several methods for model simulation and analysis, simplifying both metabolic network reconstruction and data interpretation.

- **ModelSEED** – A web-based platform for reconstructing, comparing and analyzing metabolic models. It is powered by the ModelSEED Biochemistry Database, which serves as a foundation for both ModelSEED and KBase. The platform provides mechanisms for model editing, gap-filling, and integration of experimental methods or algorithms. Although a login is required to access the platform, the project is open-source and API clients for Perl and Python are offered.

- **KBase** – A systems biology knowledge base developed by the United States Department of Energy (DOE) to advance the creation of increasingly realistic biological models. KBase is an open-source software and data platform that integrates numerous databases and tools to streamline sophisticated biological analysis. Its user-friendly Narrative Interface, based on Jupyter[10], allows users to build, store, run and share entire workflows of metabolic model reconstructions. Additionally, KBase offers a suite of scientific applications integrated to the Narrative Interface that support other research areas beyond the scope of this work.

- **MetExplore** – An all-in-one web-based platform offering an integrated suite of interactive tools for GEM exploration and curation. Its system of interactive tables simplifies data exploration and visualization. The platform offers a collaborative environment with a graphical user interface to facilitate the annotation, curation and editing of metabolic networks. MetExplore also includes functionalities for performing flux analysis, such as FBA, FVA, and gene knockout, consolidating many capabilities into a single, user-friendly tool.

- **Pathway Tools** – A software suite, associated with BioCyc database collection, that supports various use cases, including the development of organism-specific databases, metabolic reconstruction, metabolic-flux modeling, and pathway and biological network analysis. Furthermore, its capabilities of visualization and web publishing are used to power several websites, e.g., BioCyc PGDBs (Pathway/Genome Databases).

- **CarveMe** – A Python-based tool for automatically reconstructing metabolic models using a top-down approach based on a manually curated universal model. The universal model is refined by analyzing genome annotation data to identify reactions and metabolites that should be removed

[10]More information at: https://jupyter.org/about.

("carved") to produce an organism-specific metabolic model. The resulting model is simulation ready, which means it includes import/export reactions, a universal biomass equation, and no blocked or unbalanced reactions, allowing immediate use in FBA.

- **merlin** – An user-friendly Java application designed for reconstructing GEMs. It facilitates the process by retrieving data from multiple sources, providing semi-automated annotation, and assisting in model curation and validation. In addition, *merlin* includes tools and features that further support the reconstruction process.

- **mergem** – A Python package and command-line tool designed for comparing and merging metabolic models. It uses a mechanism of universal IDs, precomputed by integrating data from various biochemical databases and their cross-references, to unify and reconcile information from multiple models. The algorithm for generating these IDs can be re-executed whenever databases are updated, ensuring compatibility with new entities. Additionally, its integration with *fluxer* allows users to easily visualize the final merged model and the differences among the original models through a web interface.

- **MetaNetX** – A web platform for analyzing, comparing, searching and merging GEMs. It integrates data of models, metabolic pathways and reactions from various sources through the use of a common namespace, called MNXref, for reconciling the identifiers. The platform also offers the possibility of merging models and performing flux analysis through the web interface.

- **MEMOTE** – An automated test suite for GEMs that performs quality checks to ensure models comply with established standards and minimal requirements. It generates detailed HTML reports containing quantitative and qualitative metrics, such as the number of genes, metabolites and reactions, information about missing annotations, stoichiometric consistency, mass and charge balance, and compliance with SBML format. Additionally, its command-line interface (CLI) simplifies model versioning and the generation of reports across different versions. The CLI can also be integrated with GitHub CI to automatically produce reports whenever a new version of the model is uploaded.

- **FROG** – An ensemble of constraint-based methods used to evaluate GEMs and generate standardized, reproducible reports. It was developed as a community effort to standardize GEM evaluation and ensure reproducibility. FROG analysis is currently integrated into BioModels' workflow[11], alongside the MEMOTE test suite. The authors recommend that FROG reports be published together with curated models to support assessment and reproducibility.

- **Escher** – An open-source web-based tool for creating and sharing visualizations of metabolic pathways. It provides visualizations of prebuilt pathway maps and enables the construction of custom maps based on uploaded GEMs. Its interactive features allow seamless navigation through the metabolic network, easy retrieval of information about metabolites and reactions, and simulations of reaction knockouts and flux changes.

- **fluxer** – A web application designed to generate interactive flux graphs from GEMs based on metabolic network fluxes calculated using FBA. The tool enables users to load models from a curated dataset, upload their own in SBML format, export and share visualizations, and merge models through its integration with *mergem*.

[11] A repository of models of biological and biomedical systems. Available at: https://www.ebi.ac.uk/biomodels

- **Expasy** – The bioinformatics resource portal of the Swiss Institute of Bioinformatics (SIB). It provides access to hundreds of databases and tools developed by SIB groups, which supports life sciences and clinical research domains. Expasy also offers a user-friendly search engine that simplifies queries across multiple databases and resources, which contributes to data findability and accessibility.

Table 1.2: Tools commonly used in GEM reconstruction and their core features.

Tool	URL	Core features
The COBRA Toolbox [40]	https://opencobra.github.io/cobratoolbox	Analysis, Simulation
COBRApy [39]	https://opencobra.github.io/cobrapy	Analysis, Simulation
CNApy [58]	https://cnapy-org.github.io/CNApy	Analysis, Simulation
eQuilibrator [59]	https://equilibrator.weizmann.ac.il	Thermodynamic analysis
The RAVEN Toolbox [31]	https://sysbiochalmers.github.io/RAVEN	Reconstruction, Analysis
ModelSEED [28, 60]	https://modelseed.org	Reconstruction, Analysis
KBase [61]	https://www.kbase.us	Reconstruction, Analysis
MetExplore [62]	https://metexplore.toulouse.inrae.fr	Reconstruction, Analysis
Pathway Tools [63]	https://bioinformatics.ai.sri.com/ptools	Reconstruction
CarveMe [27]	https://github.com/cdanielmachado/carveme	Reconstruction
merlin [64]	https://merlin-sysbio.org	Reconstruction
mergem [9]	https://github.com/lobolab/mergem	Merging, Comparison
MetaNetX [36]	https://www.metanetx.org	Standardization, Annotation
MEMOTE [43]	https://memote.readthedocs.io	
FROG [65]	https://www.ebi.ac.uk/biomodels/curati	
Escher [66]	https://escher.github.io	
fluxer [67]	https://fluxer.umbc.edu	
Expasy [68]	https://www.expasy.org	

Disambiguação de dados, inconsistências e integração de diferentes fontes são os maiores desafios na reconstrução de GEMs. Gerar e/ou selecionar perguntas que explorem tais desafios parece ser um caminho que trará resultados úteis.

1.3 Key Challenges in GEM Reconstruction

Although various efforts have been made over the years to establish representation standards for biochemical data, a universal consensus has not yet been reached [9]. Challenges span the representation of metabolites, reactions, metabolic pathways, and ultimately, the entire metabolic network. The lack of universal representation standards, combined with the vast number of datasets, hinders the development of automated tools for GEM reconstruction, as these tools must handle data disambiguation, resolve inconsistencies, and integrate all information into a unified model.

1.3.1 Data Representation Issues

Names and identifiers do not follow a unified standard. As reported by *Pham et al.* [11], many popular biochemical databases contain internal ambiguity (one name pointing to multiple IDs) and multiplicity (a single ID linked to multiple names). In some cases, multiplicity can be positive, as it helps identify synonyms in the dataset, for example, KEGG compound C00001 is associated to the names “H20” and “Water”. However, when combined with ambiguous data, it can lead to spurious relationships and cause serious inconsistencies, both internally (within a dataset) and externally (among datasets).

Chemical structures, such as those of metabolites, are typically represented using string-based formats, such as SMILES (Simplified Molecular Input Line Entry System) [69] and InChI (International Chemical Identifier) [70], which are compact and easily searchable. Despite being widely adopted, these formats still present notable limitations. While SMILES strings offer a human-readable format for representing chemical structures, they are not suitable for all molecular types – specific types of polymers, for example, cannot be adequately encoded. Additionally, SMILES can introduce ambiguities, as multiple valid strings may represent the same molecule [6] – for instance, a water molecule can be represented as [O], [OH2], or [H]O[H]. Some of these ambiguities are mitigated by the InChI system, which offers a more structured and detailed representation of molecular structures. Nevertheless, InChI also has limitations: some molecules encodable by SMILES may lack a proper or complete InChI representation, and the optional nature of certain InChI fields can lead to incomplete descriptions in specific contexts [6]. Figure 1.2 shows how a single molecule can be represented in multiple ways using these formats.

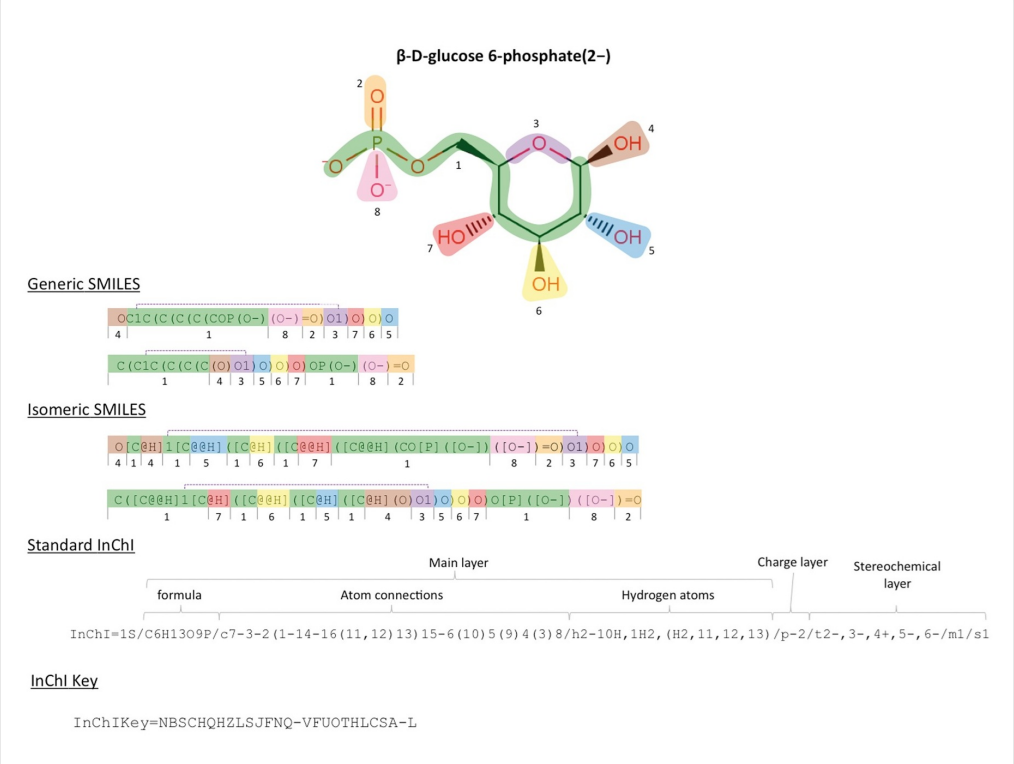


Figure 1.2: Illustration of various valid representations of a single molecule, first using SMILES strings and then the InChI system. Source: [6].

Although these formats are widely adopted, some datasets use alternative standards to represent 2D and 3D molecular structures, for example, MOL [71], KCF (KEGG Chemical Function) [72], KCF-S (KEGG Chemical Function and Substructure) [73]. The MOL format encodes molecular structures as a connection table, specifying atoms, bonds, spatial coordinates, and other molecular attributes. Alternatively, KCF and KCF-S represent biochemical structures as graphs, where atoms are nodes, bonds are edges, and substructures are defined by additional attributes. While MOL files can be difficult to interpret due to their low-level structure, KCF and KCF-S are designed to be more intuitive for biochemists by providing explicit mappings between chemical names and their computational representations.

Expert decisions play a crucial role in metabolite representation, whether by selecting the most

suitable format for a given domain or by omitting structural aspects that should be included in the dataset. *Bernard et al.* [6] summarize the various challenges involved in reconciling metabolites from different sources (Figure 1.3), underscoring the importance of careful decision-making in how metabolites are represented.

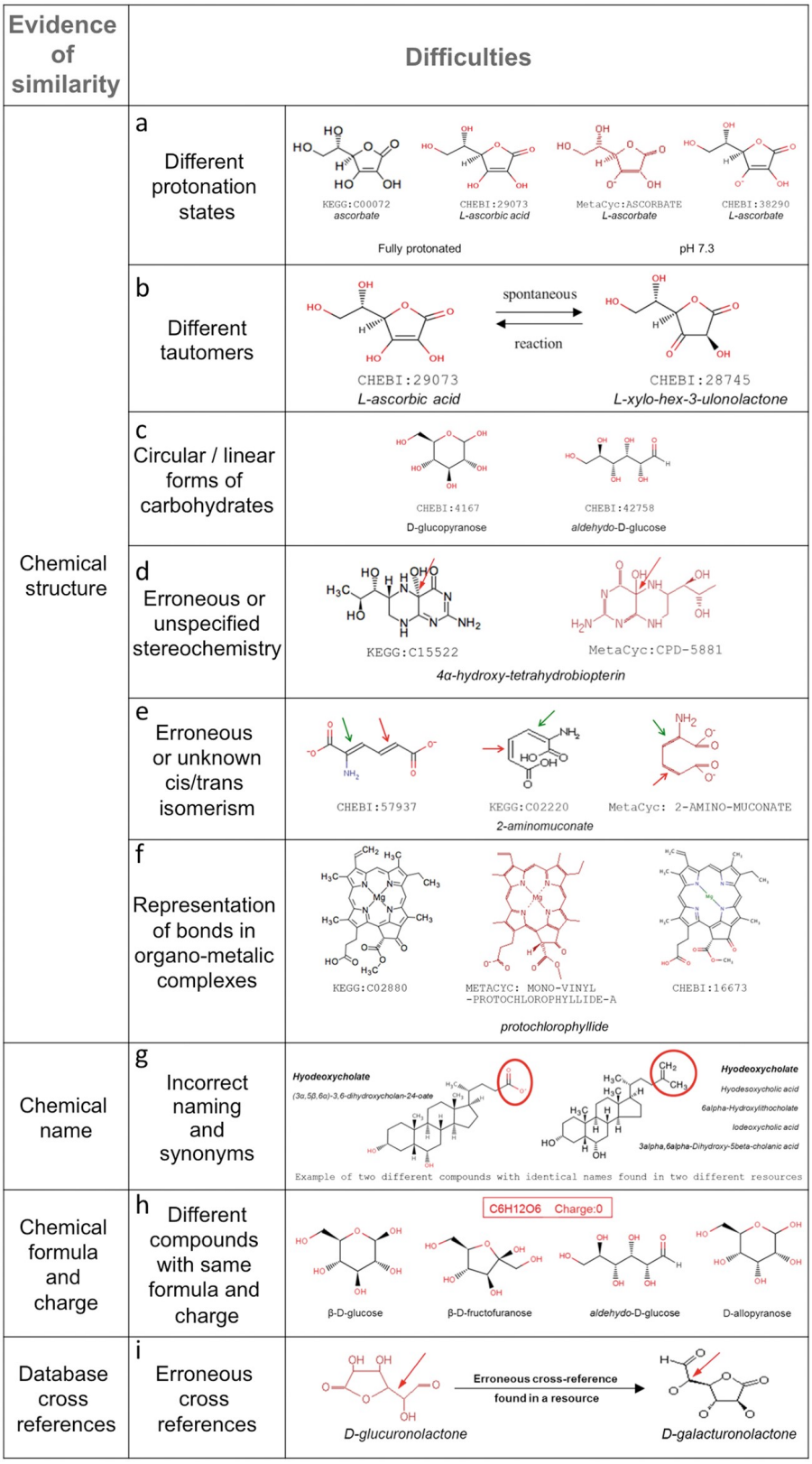


Figure 1.3: Common issues in metabolite reconciliation, caused by errors or differences in representation. Source: [6].

Since reactions represent the transformation of metabolites (i.e., the conversion of substrates into products), their representation is inherently affected by issues in metabolite representation. Furthermore, the diversity of formats used to describe reactions – ranging from simple chemical equations (such as $R1 : A + B \rightarrow C$, where A, B and C are metabolites) to more complex representations such as reaction graphs, hypergraphs, or stoichiometric matrices (Figure 1.4) [5, 23, 74, 75] – adds another layer of complexity to the reconciliation of biochemical data.

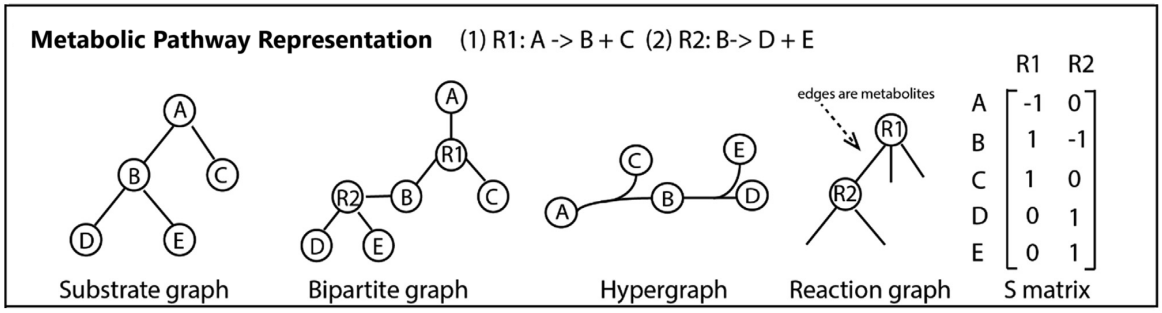


Figure 1.4: Possible representations for a metabolic pathway containing two reactions R1 and R2: a reaction list, different graph types, and a stoichiometric matrix. Adapted from [5].

Chemical equations lack critical details about enzymes, mass and charge balance, and environmental or organism-specific characteristics that may influence the reaction. Moreover, representing reactions through formulas is susceptible to nomenclature-related issues, similar to those encountered in metabolite representation. As illustrated in Table 1.3, a single reaction can be expressed in various ways, depending on the naming conventions adopted by different research groups or datasets [76].

Table 1.3: Multiple representations for the reaction *glucose-6-phosphate dehydrogenase* found in scientific literature and datasets. Adapted from [76].

Reaction – *glucose-6-phosphate dehydrogenase*
`[c]: g6p + nadp < = = > 6pgl + h + nadph`
`[c]: g6p + nadp - > 6pgl + h + nadph`
`[c]g6p + nadp < = = > 6pgl + h + nadph`
`[c]: f420-2 + g6p - > 6pgl + f420-2h`
`G6P + NADP < - > D6PGL + NADP`
`G6P + NADP - > D6PGL + NADPH`
`G6P + NAD - > D6PGL + NADH`
`C01172 + C00006 = C01236 + C00005 + C00080`
`C00092 + C00006 < = > C01236 + C00005 + C00080`

In addition, metabolic networks and models comprise more than just a list of reactions. In GEMs, each reaction is associated with genes and enzymes through gene-protein-reaction (GPR) rules, along with other attributes essential for simulating the organism's behavior. SBML remains the dominant format for representing GEMs, as its design facilitates model exchange and reuse while supporting simulation-related properties [77]. However, SBML lacks a formal specification for mandatory fields, allowing different tools to populate fields inconsistently and potentially leaving some incomplete or filled with inappropriate information [12]. Moreover, GEM reconstructions may vary due to differences in methods, algorithms and expert decisions throughout the process, even when using same data sources [35] or rule-based methods (e.g., GPRuler [78]).

Since string-based methods alone cannot account for all these variations, more sophisticated techniques are required to reconcile information about metabolites and reactions. Some approaches, such as MNXref [6] and MetRxn [76], address this challenge by integrating data from multiple biochemical datasets through iterative processes that leverage nomenclature and cross-references to resolve ambiguities and provide a unified, reliable data source. However, these methods may overlook critical aspects such as reaction directionality, compartmentalization, mass balance, and stoichiometric coefficients, which can potentially lead to inaccurate results. Moreover, the widespread presence of internal and external inconsistencies in names and identifiers across biochemical databases further complicates the reconciliation process [11].

Some of the tools mentioned earlier aim to automatically reconstruct GEMs by integrating data from multiple sources. However, most of them fail to generate models that are immediately usable for simulation [79]. A recent study [80] reports that among the most widely used reconstruction tools, only CarveMe, ModelSEED, and *gapseq* generate models ready for flux analyses. Even so, these "ready-to-use" models are only initial drafts that need to be manually validated and refined by experts to ensure reliability. Additionally, most existing tools and methods are evaluated within a limited scope, typically using well-studied organisms such as *Escherichia coli*, which restricts their applicability to other organisms or environments [79].

Given these challenges, many researchers consider expert manual curation as an indispensable part of GEM reconstruction. As a result, rather than pursuing fully automated solutions, they have developed hybrid approaches to support and simplify the curation process [9]. Tools such as ModelSEED, KBase, and *mergem*, for example, offer user-friendly interfaces for model refinement while automating parts of the reconstruction workflow. Such approaches represent a promising direction, as they help reduce the manual workload while preserving model accuracy and enabling the development of high-quality GEMs.

1.3.2 Experimental and Literature-based Model Validation

Experimental evidence, whether obtained from biological experiments or scientific literature, is essential for validating GEM reconstructions and ensuring their reliability. Since biological experiments are not always feasible, literature review is often the primary method for gathering this evidence [26]. Organism-specific books and scientific articles can provide insights into the presence and function of genes, enzymes, and proteins [44], and contribute to the identification of issues in the model, such as missing reactions, incorrect substrate or cofactor specificity, unbalanced reactions, incorrect reaction directionality, and erroneously added metabolites or reactions. If left unresolved, many of these issues can significantly affect the accuracy and predictive power of the *in silico* model. Additionally, data extracted from literature are often more heterogeneous than those from biochemical databases, posing additional integration challenges [11].

Although tools such as PubChem, facilitate literature-based data integration by linking chemical entities and terms to scientific publications [81], a significant portion of this information is manually added by authors or publishers, who associate relevant terms with PubChem entities. References are automatically incorporated in PubChem using MeSH (Medical Subject Headings) annotations, which are commonly found in scientific papers. While such tools aid the literature search process, experts must still validate the retrieved information to ensure it provides strong and relevant experimental evidence for their research goals.

1.3.3 Research Fragmentation and Collaboration Barriers

Although researchers disseminate their findings through publications and conferences, they do not always collaborate as a unified community with a shared goal. Research groups are usually affiliated with different universities or companies (sometimes in different countries), which may have distinct funding sources, research objectives, time constraints and scope limitations [82]. These research silos are a key factor behind the proliferation of fragmented solutions, such as the numerous tools and databases with overlapping functionalities discussed earlier. As a result, the abundance of alternative solutions further hinders the adoption of universal standards for building and sharing GEMs.

Some studies indicate that experts often choose to reconstruct models from scratch due to limited awareness or familiarity with existing computational resources [5, 24]. Poor documentation and counterintuitive interfaces also discourage the adoption of publicly available tools and data standards. BioPAX, for example, is an OWL-based standard developed for representing and sharing biological pathway data, but its adoption has been relatively limited compared to SBML. A recent report from the BioPAX team [83] highlights the main reasons for this: despite the availability of several tools supporting the format, the standard remains "poorly used by the scientific community" due to the lack of user-friendly interfaces and the absence of crucial features, such as direct knowledge reasoning and extraction capabilities, which are highly desirable for ontology-based resources like BioPAX. Overcoming these usability barriers and minimizing the learning curve are essential steps for increasing the adoption of data standards and encouraging broader use of databases and tools.

Developing a universally adopted solution requires a collaborative effort to understand and accommodate the specific needs of different research groups [82]. Without effective collaboration, teams may fail to recognize or address each other's requirements. This effort must involve both life science experts and computer scientists with expertise in software development, data management, and computing infrastructure, as a team limited to only one of these domains is unlikely to achieve sustainable and broadly useful outcomes. Therefore, initiatives that foster interdisciplinary collaboration and knowledge exchange should be actively promoted [24].

Chapter 2

Ontologies and Semantic Interoperability

As discussed in the previous chapter, many challenges in GEM reconstruction stem from the lack of standardized representations for biochemical data and their fragmentation across multiple sources, which hinders the implementation of automated solutions. Ontologies offer a way to address this issue by standardizing knowledge representation through the explicit definition of concepts and their relationships, thereby facilitating the reconciliation of information exchanged between systems. This chapter explores the role of ontologies in enabling system interoperability and discusses their application in GEM reconstruction.

2.1 Ontologies

Ontology is a branch of philosophy that studies the nature of existence, what things are, and how they can be described, classified, and how they relate to each other [84]. Ontologies provide semantic meaning by linking symbols to real-world entities. Rather than establishing a single, final definition of an entity, an ontology determines what should be considered true within a given domain, shaped by a specific worldview, which is also known as an **ontological commitment** [85].

In computer science, ontologies are a key component of **Knowledge Representation**, a field that focuses on structuring information in a machine-readable format to support automated reasoning and decision-making. Ontologies can be categorized according to their purpose and scope [86]. **Upper ontologies** (also known as **foundational ontologies**) offer a structured set of domain-independent ontological theories, they represent general aspects of reality by defining high-level, abstract concepts such as time, space, objects and processes. **Domain ontologies**, on the other hand, focus on the representation of concepts and relations within a specific domain, for example, chemical substances, molecules, and reactions. Lastly, **application ontologies** can be interpreted as an extension of domain ontologies, tailored for specialized environments or software applications.

To facilitate the representation of complex concepts and their relationships, the World Wide Web Consortium (W3C) developed a Semantic Web language named **Web Ontology Language (OWL)** [87]. OWL is a logic-based language built on the **Resource Description Framework (RDF)** [88], which represents relationships between entities using subject-predicate-object triples. OWL follows an open-world assumption, which means that unless a fact is explicitly declared, it cannot be assumed to be true or false. Moreover, it allows the definition of complex statements through axioms and expressions, which enables richer and more sophisticated representations. The OWL syntax alone does not ensure a representation adheres to ontological assumptions, as its purpose is only to provide

the structure needed to build ontology-based representations [17].

Due to the logical properties and constraints of these representations, **reasoning algorithms (reasoners)** can be implemented to verify data consistency and uncover implicit knowledge through logical inferences [87]. By making the meaning of each concept explicit and enabling the tracing logical inferences, ontology-based representations also support system explainability and interpretability [89]. Moreover, explicit definitions contributes to interoperability by minimizing ambiguities and facilitating the mapping of terms and concepts in different systems [17, 84, 85].

2.1.1 Ontology Development Methodologies

The process of building an ontology involves more than simply creating a representation for defining and mapping terms; it also requires establishing consensus among various stakeholders (e.g., domain experts, system users) who interact with this information [90]. Several methodologies have been proposed to provide systematic frameworks for ontology development, such as METHONTOLOGY [91], Ontology Development 101 [92], NeON [93], SABiO [94], and OntoForInfoScience [95]. Although these methodologies differ in specific details, most of them share the same core steps (Figure 2.1), as described below:

1. **Specification** – Identify the purpose of the ontology and define its scope. Many methodologies recommend formulating Competency Questions (CQs) [96] to specify the types of queries the ontology should be able to address, which can also serve as a basis for its subsequent evaluation.

2. **Conceptualization** – Create an informal representation for the knowledge using diagrams or natural language descriptions. At this stage, knowledge is organized by defining key concepts, relationships, properties, and constraints.

3. **Formalization** – Develop a formal representation by defining classes, individuals, properties, and logical constraints. Although the formal model in this stage is independent of a specific syntax, it often relies on Description Logic concepts. This model must be theoretically sound, consistent, and unambiguous.

4. **Implementation** – Encode the ontology in a machine-readable formal language, such as OWL. Ontology development tools, such as Protégé[1], can be used in this phase to simplify the process.

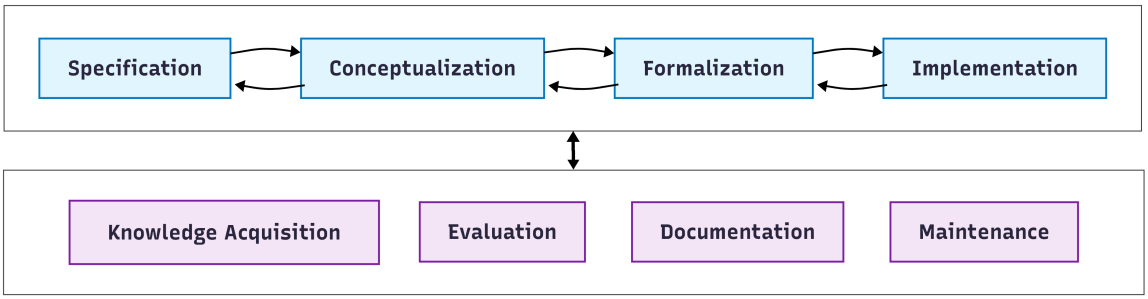


Figure 2.1: Ontology development core steps and supporting activities.

Ontology development follows an iterative approach, where the steps are repeated multiple times until the final version is achieved. Steps 3 and 4 are often closely intertwined, as ontology development

[1]Available at: https://protege.stanford.edu/

tools provide graphical interfaces that facilitate the construction of formal representations and enable consistency validation through automated reasoning. Throughout this process, competency questions can also be refined and expanded as the domain knowledge grows and new requirements emerge.

Most methodologies additionally propose a few supporting activities that should be conducted in parallel with the core development steps; the most important are:

- **Knowledge acquisition** – Collect information from domain experts, scientific literature, databases, documentation, and other relevant sources. Analyze existing ontologies for potential reuse. After defining the core structure of the ontology – including classes, relationships, and axioms – automated scripts are used to extract data from specified datasets and populate the ontology with entities (individuals).

- **Evaluation** – Perform ontology verification (checking correctness, consistency, and clarity of representation) and validation (ensuring completeness, adherence to scope requirements, and competency questions). While automated tools can support evaluation, domain expert involvement is essential in this activity.

- **Documentation** – Provide a comprehensive documentation for the ontology, including its structure, classes, relationships and properties. This documentation should also explain design decisions and provide guidelines for usage and maintenance.

- **Maintenance** – Establish strategies and procedures for publishing, updating and versioning the ontology. Ideally, the ontology should be extensible to encourage reuse; therefore, participation from the broader community in extending the ontology, resolving issues, and submitting requests should be considered and actively supported.

2.1.2 Ontologies for Biology and Chemistry

Ontologies have proven to be powerful tools for addressing data integration and system interoperability challenges across various domains, including chemistry and biology [28, 85, 86]. The Basic Formal Ontology (BFO) [97], for instance, is a widely recognized upper ontology that provides foundational concepts for building domain-specific ontologies, such as ChEBI [19] and GO [18], which are also among the most popular ontologies in their respective domains. Due to its importance, BFO fostered the creation of The Open Biological and Biomedical Ontology Foundry (OBO Foundry), a collaborative community that has established principles and best practices for developing interoperable ontologies in life sciences.

The ChEBI (Chemical Entities of Biological Interest) is among the most popular ontologies in chemistry. It represents chemical molecules (e.g., caffeine, methanol, etc.), their properties (ID, definition, formula, SMILES, etc), and their hierarchical categorization (e.g., methanol is categorized as an alcohol and an organic molecular) [38]. The Gene Ontology (GO) [18] provides a comprehensive representation of gene functions and products. Its structure is organized into three aspects: Molecular Function – molecular activities performed by gene products (e.g., catalysis), Cellular Component – cellular locations and structures (e.g., cytosol), and Biological Process – processes composed of interconnected molecular activities (e.g., DNA repair). Both ChEBI and GO identifiers are widely used to annotate databases, providing cross-references that minimize ambiguities and enrich the dataset.

The ontology most directly associated with GEM reconstruction is the Systems Biology Ontology (SBO) [20], which defines terms related to the domain, including physical entities (e.g., metabolites,

genes, biomass) and processes (e.g., biochemical reactions). The close association between SBO and GEMs even led to the inclusion of a specific attribute for SBO identifiers in the SBML specification, facilitating the integration of models originated from different sources [98]. Despite this integration, SBO does not adequately capture the complex relationships between entities, which limits its effectiveness in representing the full structure and semantics of GEMs. Additionally, no existing ontology captures this level of detail in GEM representation, highlighting the potential of a dedicated ontology as an innovative solution to overcome these limitations.

2.2 Semantic Interoperability

The volume and complexity of data produced by research groups worldwide have increased rapidly over the years. However, this growth has not always been accompanied by the adoption of standards that enable data discovery, integration, and reuse across diverse sources. For example, when comparing or integrating data from different studies, researchers often struggle to locate relevant information, determine its availability for download, assess its compatibility with their own datasets, and verify whether permission is required for access and use. In order to address these issues, *Wilkinson et al.* [99] proposed the **FAIR (Findability, Accessibility, Interoperability, and Reusability) Data Principles**, a set of guidelines aimed to improve data management, stewardship and sharing across various teams and computational systems.

According to these principles, to ensure interoperability, systems must use a formal, accessible, shared, and widely applicable knowledge representation language that supports terminological alignment and simplifies data access. However, while these recommendations enable interoperability at the levels of **syntax** (through the use of a common vocabulary) and **communication** (by ensuring information is accessible), they fall short of achieving **semantic interoperability** [17]. This level of interoperability requires preserving the ontological commitments of each system and accurately capturing various types of relationships between entities (e.g., identity, specialization, historical dependence). Therefore, adopting a semantics-driven representation approach, grounded in ontologically consistent decisions, is essential for implementing semantic interoperability, as it ensures consistency between the knowledge base and the domain it represents and promotes stronger alignment with the FAIR principles [17].

The OWL format alone does not guarantee that a representation contains rich semantic information. Although many ontologies in the biochemistry domain (e.g., ChEBI, GO, SBO) provide a common vocabulary and a hierarchical classification of entities that is ontologically consistent, they often lack detailed representations of the relationships between those entities. As a result, integration between biochemical databases and systems is typically limited to the levels of syntax and communication. This reinforces the hypothesis that a semantically enriched ontology for GEMs could offer significant benefits for improving interoperability across specialized databases and tools used in GEM reconstruction.

2.2.1 Enabling Interoperability with Ontologies

In healthcare and life sciences, ontologies have been applied in a variety of contexts, with numerous cases demonstrating their role in enabling system interoperability. For example, in the Brazilian public healthcare system, data on pregnancy, birth rates, and infant mortality were distributed across multiple databases managed by different stakeholders. This fragmentation often led to problems when executing complex queries that required information from various sources. To address this issue, an

Ontology-Based Data Integration (OBDI) approach was adopted, using a layer of ontologies to integrate the databases and centralize queries through a single access point [14, 100].

A similar use case in Brazil, involving the integration of systems used to monitor *tuberculosis* [13] cases, was also addressed through an ontology-based approach. Data on patients, hospitalizations, exams, and treatments were partially stored in legacy systems that needed to be maintained due to operational constraints. To enable integration without significant system changes, an ontology layer was developed to map terms and centralize system access. This layer was grounded in BFO (Basic Formal Ontology) [97], a foundational ontology, to support term mapping, facilitate interoperability with other BFO-based ontologies, and promote ontologically consistent decisions throughout the development process.

Data expressivity and interpretability can be improved through a technique known as ontological unpacking, which clarifies the meaning of concepts by explicitly defining entity types and relationships [85]. This approach reduces ambiguity and facilitates the identification of various types of relationships between entities across different systems, thereby streamlining the implementation of semantic interoperability. Recent studies [16, 101] have demonstrated promising results from applying this technique in the field of biology, particularly in facilitating information exchange among teams working with similar datasets. These successful use cases suggest that ontologies can offer an effective solution to the interoperability challenges encountered in GEM reconstruction. The next chapter will provide further details on this approach.

Chapter 3

Research Proposal: An Ontology for GEMs

Developing an ontology specialized in representing GEMs could help overcome interoperability challenges in their reconstruction and make the process more efficient. This chapter begins by presenting a few motivating scenarios for the construction of a GEM ontology. It then describes the methodology adopted for ontology development and outlines the proposed strategy for evaluating the overall quality of the solution. Finally, the chapter discusses the main challenges and limitations of the proposed approach.

3.1 Motivation and Expected Contributions

Although the use of ontologies in the fields of chemistry and biology is well-established, their potential to enable interoperability across biochemical databases remains underexplored, creating opportunities for novel research and innovative computational solutions. By creating a comprehensive and semantically enriched metabolic model, we aim to establish a unified representation for GEMs, thereby minimizing the data inconsistency issues previously described. Nomenclature discrepancies, for instance, can be addressed by integrating the proposed ontology with widely recognized databases such as ChEBI. This integration could help recognize and disambiguate generic compound names commonly used to identify reaction substrates and products. A similar strategy applies to enzymes: the use of E.C. numbers can help standardize enzyme identification and link reactions to their associated catalytic functions.

Semantic rules can also be used to validate model construction. For example, when a reaction occurs within a given compartment, its substrates and products are expected to reside in the same compartment (except for transport and exchange reactions). Incorrect compartment assignments can distort FBA results; however, such constraints are neither enforced by SBML nor validated by current tools, making their detection a time-consuming task. Implementing these validations within the ontology is relatively straightforward and could substantially reduce the manual effort required by domain experts to identify and resolve such issues.

The use of an ontology is also expected to enhance the explainability of both the model and the reconstruction process. Existing automated reconstruction tools often lack mechanisms to clearly justify their outputs. For instance, while MetaNetX provides useful reports on unification issues, its documentation is limited, and the algorithms behind its reconciliation processes are not publicly available. On the other hand, although *mergem* shares its source code, it is poorly documented and difficult to interpret – even for experienced software developers. Furthermore, *mergem* does not provide de-

tailed reports on its reconciliation results, which undermines the transparency and trustworthiness of its outputs. An ontology-based approach can offer traceable, interpretable results and support the generation of explainable reports about the reconstruction process.

The development of the proposed ontology-driven solution is expected to yield important contributions to the field of Computer Science, including:

- **Identifying domain-specific challenges** – Developing an ontology-based application to support GEM reconstruction may uncover domain-specific challenges not addressed in existing studies on ontologies and interoperability. Throughout this work, we aim to identify these challenges and propose novel solutions to overcome them.

- **Designing an interoperability framework for GEM reconstruction** – Existing solutions do not offer a fully integrated environment to facilitate metabolic model reconstruction. In parallel with the ontology development, we plan to design an architecture that will serve as a foundation for future integrated solutions, along with a proof of concept containing the core functionalities of this framework. This architecture is expected to guide the development of a fully integrated solution in the future.

- **Simplifying application development** – An ontology-driven approach shifts complexity from the application layer to the knowledge base by embedding semantic information and logical structures within the ontology. A domain-specific ontology facilitates data unification, supports advanced search capabilities, and provides a reliable foundation for knowledge representation. As a result, applications built on top of the proposed GEM ontology are expected to be simpler, more maintainable, and more extensible, potentially fostering the development of additional tools to support GEM reconstruction in the future.

- **Enabling automated GEM repairs** – Representing GEMs with ontologies opens the possibility of applying ontology repair techniques to automatically resolve inconsistencies in the models. Although the development and application of such techniques lie beyond the scope of this work due to their complexity, the outcomes of this project will serve as a foundation for future research in this direction.

Furthermore, this ontology-driven approach is anticipated to deliver the following key contributions to Systems Biology:

- **Accelerating the GEM reconstruction process** – A considerable amount of time is currently devoted to reconciling information from multiple sources and resolving inconsistencies in the metabolic model. By streamlining these tasks through an ontology-based approach, manual curation efforts can be reduced, thereby accelerating the overall reconstruction process.

- **Facilitating knowledge sharing and model reuse** – The explicit definition of concepts and their relationships is expected to enhance model explainability and support effective knowledge sharing across teams. In addition, the use of logical inferences to disambiguate concepts, detect inconsistencies, and reconcile information enables traceability of automated decisions, thereby improving interpretability. Together, these features are expected to facilitate the sharing and reuse of GEMs.

- **Supporting knowledge discovery** – The logical capabilities of the ontology are expected to enable automated reasoning, allowing for the detection of inconsistencies and the the extraction of meaningful knowledge from the GEMs. As a result, the application is anticipated to provide deeper insights into the models and potentially lead to the discovery of novel aspects of the organism's metabolism.
- **Reducing GEM reconstruction costs** – By minimizing manual effort and accelerating the GEM reconstruction process, this project aims to lower the overall cost of generating metabolic models. This cost reduction may promote broader adoption of GEMs in diverse scenarios and potentially contribute to new scientific discoveries.

3.2 Ontology Development

The GEM ontology will be primarily used by Systems Biology experts, and it is desirable that it can be extended or updated in the future, even without the involvement of professional ontologists. Therefore, a key requirement is ease of maintenance, which depends on clear and comprehensive documentation that includes usage and maintenance guidelines. Rather than strictly following a single methodology, we adopted the core steps common to most ontology development approaches, along with the supporting activities previously described. This flexible approach allows us to preserve the fundamental principles of ontology engineering while avoiding the constraints of rigid procedures, making the process more agile.

Collaboration with domain experts is crucial for the successful development and adoption of the proposed ontology. In the early stages (specification and conceptualization), domain experts actively contributed to defining the scope, eliciting requirements, formulating competency questions, and validating proposed concept and relationship descriptions. Their continued involvement will also be essential in later phases, particularly for evaluating the quality and usability of the resulting ontology.

The project began with scope definition, which required an extensive literature review to bridge the knowledge gap between Computer Science and Systems Biology. Under the guidance of systems biology experts, we studied the GEM reconstruction process, existing solutions, and the available databases and tools. Based on this foundation, the project's scope and objectives were established, and a set of competency questions was formulated to guide the domain conceptualization.

3.2.1 Competency Questions

The competency questions were formulated based on activities typically performed manually by domain experts during GEM reconstruction. These activities generally involve comparing metabolites and reactions, as well as evaluating the metabolic network (e.g., metabolite production, energy consumption, biomass generation). A top-down approach was adopted, starting with broad questions and progressively refining them into more specific ones. The final set of questions is listed below:

Metabolites

1. Are metabolites *M1* and *M2* the same?
2. Is metabolite *M1* from model *MOD1* the same as *M2* from database *DB1*?
3. Is metabolite *M1* from database *DB1* the same as *M2* from *DB2*?

4. Does database *DB1* contain metabolite *M1*?
5. Does metabolite *M1* have the correct chemical formula?

Reactions

1. Is reaction *R1* mass- and charge-balanced?
2. Are reactions *R1* and *R2* the same?
3. Is reaction *R1* from model *MOD1* the same as *R2* from database *DB1*?
4. Is reaction *R* associated with a single gene or multiple genes?
5. Does gene *G* participate in more than one type of reaction?
6. What enzymes and reactions are associated with gene *G*?
7. Which reactions are catalyzed by enzyme *E*?
8. Which enzymes catalyze reaction *R*?
9. Is reaction *R* blocked?
10. Which reactions produce metabolite *M*?
11. Does reaction *R* produce metabolite *M*?

Model/Network – Topological Analysis

1. How many blocked reactions are present in the network?
2. Which reactions are blocked in the network?
3. How many dead-end metabolites are present in the network?
4. Which metabolites are dead ends in the network?
5. Which genes can be knocked out to prevent the production of metabolite *M*?
6. Which metabolites are produced but not consumed by any reaction in the network?
7. Which metabolites are required but not produced in the network?
8. Are all reactions in the network mass- and charge-balanced?
9. Do all metabolites in the network have correct chemical formulas?
10. Does the model generate biomass?

3.2.2 Conceptualization

Based on the competency questions we identified the core concepts and relationships of the ontology and designed an initial domain conceptualization (Figure 3.1). The central component of the model is the **MetabolicModel** class, which acts as an information aggregator, encompassing properties of the organism and a list of metabolites, reactions, genes, and compartments. A **BiologicalEntity** represents entities that physically exist in the real world, associated with a unique ID, a name, and a list of cross-references including external data and metadata (**CrossReference** class). The remaining classes are described below:

- **Metabolite** – A small chemical compound that participates in metabolic reactions, either as a reactant, product, cofactor, or intermediate. Each metabolite is associated with a specific compartment (e.g., glucose in the cytosol vs. extracellular glucose are distinct entities). Synonyms and cross-references are used in the model to represent its various identifiers.

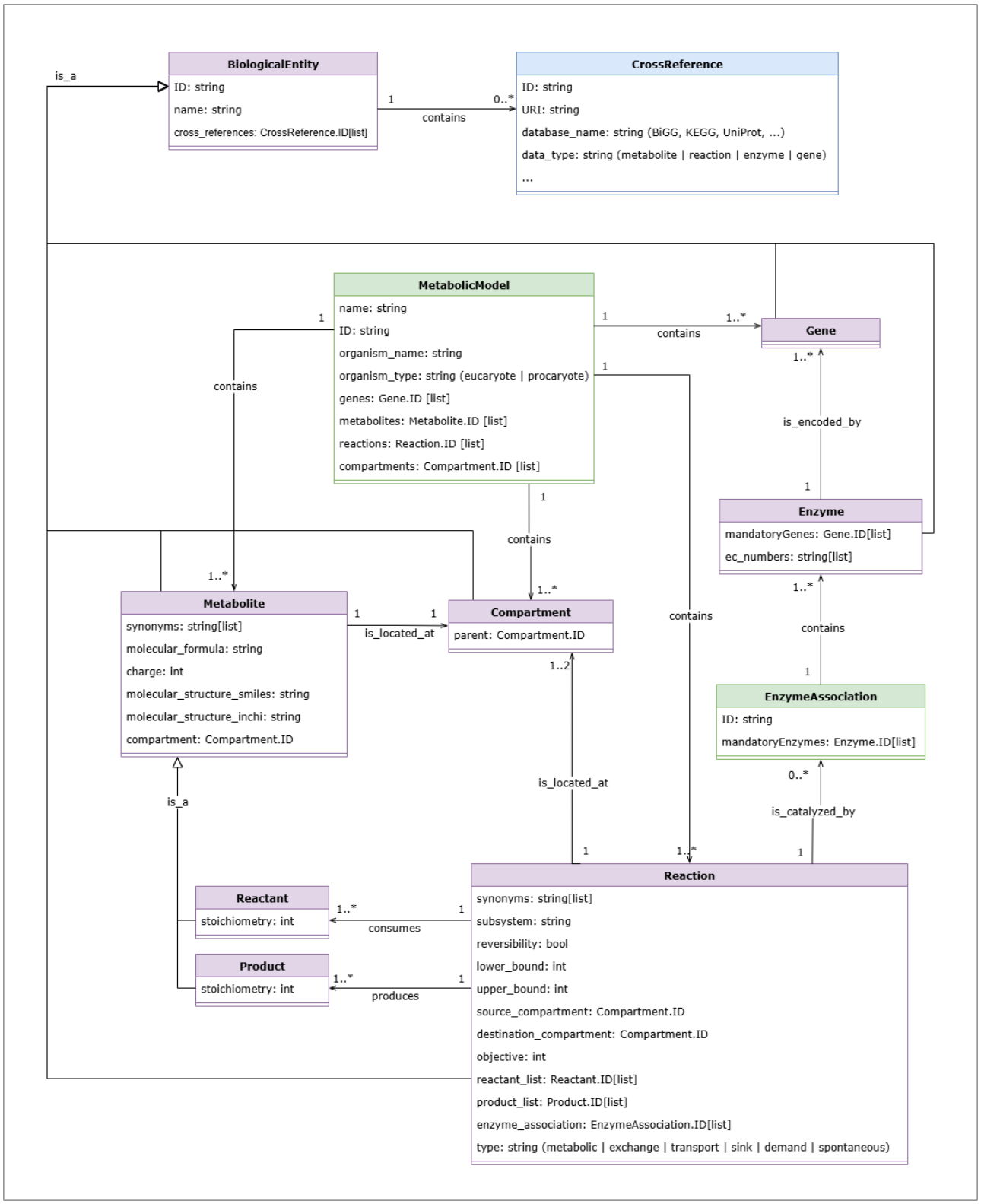


Figure 3.1: Initial domain conceptualization. Classes are represented as boxes; white-tipped arrows indicate inheritance, while standard arrows denote associations along with their cardinalities.

- **Reaction** – A biochemical transformation that converts a set of reactants into products. This class defines the stoichiometry, directionality, reversibility, subsystem (pathway), and compartmentalization of reactions, serving as a core functional unit of the metabolic network. Lower and upper bounds specify flux rate and direction, while the objective attribute indicates the reaction's role in flux optimization. Source and destination compartments are specified for transport

reactions, which move metabolites between compartments; for other reaction types, these fields typically share the same value. Enzymatic and non-enzymatic (spontaneous, transport, sink, and demand) reactions are distinguished by their relationship with the **EnzymeAssociation** class.

- **Gene** – A DNA sequence that encodes a protein, typically an enzyme involved in a metabolic reaction. Genes are connected to reactions through GPR associations, which are implicitly represented in the diagram through the relationships among the **Gene**, **Enzyme**, **EnzymeAssociation**, and **Reaction** classes. Each gene encodes either a complete enzyme or an enzyme subunit, which in turn contributes to the catalysis of one or more reactions.

- **Enzyme** – A protein or protein complex that catalyzes specific biochemical reactions within the metabolic network. Enzymes can be encoded by one or more genes, and each enzyme may be associated with one or more reactions. When multiple enzymes work together to catalyze a single reaction, they form an enzymatic complex, represented by the **EnzymeAssociation** class. This intricate relationship among genes, enzymes, and reactions represents GPR rules.

- **EnzymeAssociation** – Represents the Boolean relationships from GPR rules. It includes a list of mandatory enzymes that must act together to catalyze a reaction, representing a logical AND. Since a reaction may be catalyzed by different sets of enzymes, each reaction may have one or more instances of **EnzymeAssociation**, corresponding to a logical OR.

- **Compartment** – A region within the cell (e.g., cytosol, mitochondria, extracellular space). GEMs use compartments to localize reactions and metabolites, which allows for modeling of transport and compartment-specific pathways.

To develop the current domain model, extensive discussions were held with GEM reconstruction experts, including two researchers from ICB-USP, one from another Brazilian institution (Cidacs FIOCRUZ-BA), and one from a European institution. These discussions were fundamental to establish the project scope, elicit the requirements, formulate competency questions and build the conceptualization.

Most discussions focused on developing a clear understanding of the domain – from fundamental biological concepts such as metabolites, reactions, and enzymes to the overall GEM reconstruction process. Several resources supported these discussions and helped foster a shared understanding among the team: the competency questions guided the identification of domain expert needs; the SBML specification served as a reference for commonly used entities and properties in metabolic models; and the diagram in Figure 3.1 functioned as a visual artifact for building consensus around the meaning of entities and their relationships.

One important topic that required extensive discussion before reaching consensus was the representation of reactions. Defining the attributes of the **Reaction** class involved determining which properties could be used to uniquely identify a reaction – an essential step for reconciling reactions across different models or datasets. These discussions led to the inclusion of the **Enzyme** class in the model, an entity not explicitly defined in the SBML specification, as an additional feature to support more accurate reaction comparison. The inclusion of the **Enzyme** class required a reassessment of how GPR associations are represented. In SBML, reactions and genes are linked through a *GeneProductAssociation* element that encodes Boolean rules. In our model, however, this relationship was restructured: genes are associated with enzymes, and enzymes (or enzymatic complexes) are then linked to reactions.

Compared to the SBML format, the representation of compartments for metabolites and reactions was slightly modified. While SBML allows for defining a list of compartments and assigning them to metabolites and reactions, most models typically append a compartment suffix or prefix to the identifiers. This practice stems from SBML's requirement that all identifiers must be unique, which prevents the same metabolite or reaction from being represented in multiple compartments without altering its ID. In our model, we removed this restriction, allowing multiple instances of the same metabolite or reaction to coexist as long as they are assigned to different compartments.

3.2.3 Formalization and Semantic Enrichment

The domain conceptualization already provides a solid foundation for anyone interested in working in this domain, as it clarifies the meaning of the most important GEM concepts. However, a semantically enriched version of this conceptualization could further improve concept disambiguation, promote explainability, and facilitate interoperability across systems. Ontological unpacking techniques, combined with the OntoUML modeling language, have shown promising results in enhancing concept definitions and making implicit knowledge explicit [16, 85, 101]. Therefore, applying these techniques to the current conceptualization may lead to a more comprehensive and semantically rich representation of GEMs.

Figure 3.2 shows a partial extension of the conceptual model, focusing on the relationship between **Reaction** and **Metabolite**, and demonstrates how this method can significantly improve the semantic representation of the model. The diagram adopts OntoUML notation to clearly differentiate the types of classes represented. In this version, **Reaction** is explicitly modeled as a *«relator»* capturing the transformation of substrates into products. This explicit representation enables a clearer definition of constraints that ensure reaction balance, i.e., consistency between substrate and product molecules. Additionally, the explicit definition of reaction roles facilitates the differentiation of properties and constraints associated with each role.

Further work is needed to achieve a comprehensive model representation. By applying ontological unpacking techniques to the entire ontology, we expect to produce a more expressive representation of GEMs, resulting in more effective integration across systems.

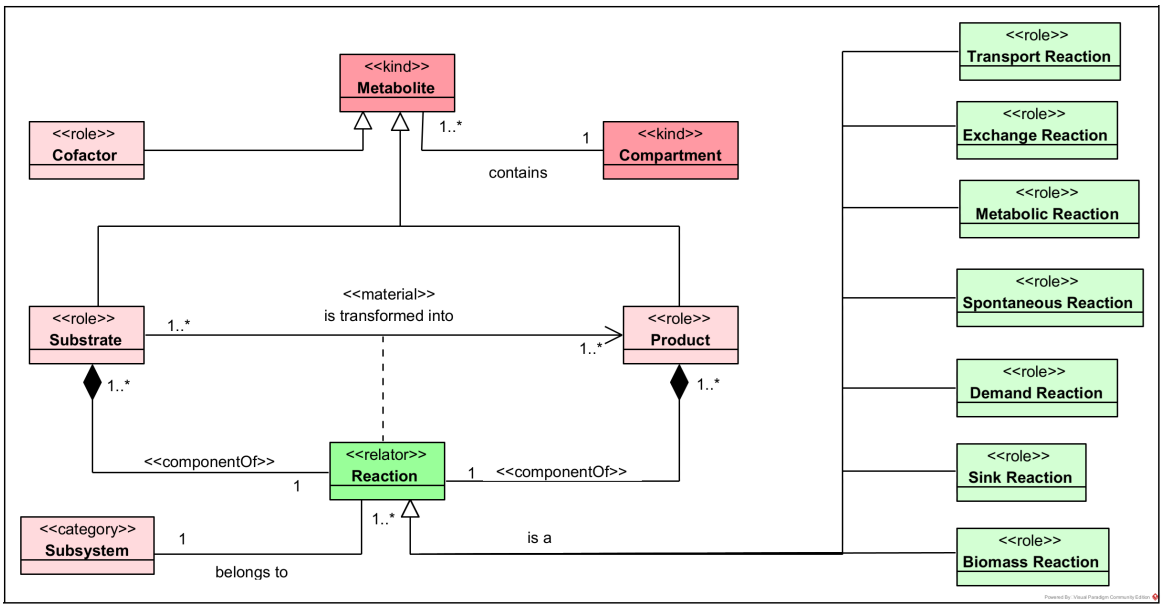


Figure 3.2: Extended version of the relation between **Metabolite** and **Reaction** entities.

3.3 Ontology-Based Application

With the GEM ontology in place, a software application will be developed to integrate it into the GEM reconstruction process. The application must be capable of loading a draft model (any valid SBML file) and automatically retrieving data from biochemical databases (e.g., BiGG, KEGG, MetaCyc, MetaNetX) and relevant ontologies (e.g., Gene Ontology, ChEBI) to enrich the model and build a standardized representation. The ontology will provide a formal structure for the data and facilitate logical consistency verification through the use of logical inferences.

The application must be able to generate a report summarizing the reconciliation results – logging automated decisions, describing unresolved issues, providing qualitative and quantitative model evaluations, and presenting additional information to assist domain experts in refining the model, either manually or automatically, depending on the nature of the issues detected. In addition, the application should enable experts to edit, visualize, and export the model in both OWL and SBML formats. The resulting model will be an enhanced version of the draft, enriched with information from biochemical databases and validated by domain experts.

The technical details of the proposed application will be defined during the implementation phase, based on the specific requirements identified throughout development. This includes selecting the most appropriate biochemical databases and ontologies to support data enrichment, defining the mechanisms for data access (e.g., APIs, SPARQL endpoints), and establishing strategies for data management and storage. Furthermore, careful planning of the application's deployment and documentation will be essential to ensure long-term maintainability, reproducibility, and usability for future users and developers.

3.4 Evaluation of Ontology-Based Reconstruction

The primary objective of the evaluation is to assess the effectiveness of the proposed ontology in improving the GEM reconstruction process. To achieve this, the evaluation must verify the ontology's consistency and correctness, as well as determine whether it meets the defined scope requirements and accurately represents the metabolic model.

The consistency of the representation can be verified through the use of automated reasoners, which are typically included in ontology development tools such as Protégé. Reasoners can identify logical inconsistencies in the knowledge base – for example, contradictory statements such as a reaction defined as occurring in one compartment while its metabolites are located in another, or two entities declared equivalent but having different property values. Currently available tools are unable to perform this type of evaluation, as SBML-encoded GEMs lack the semantic richness and reasoning capabilities provided by ontologies. The more complete the ontology, the more inconsistencies the reasoner will be able to detect.

Competency questions will be used to verify whether the ontology can answer the queries proposed by domain experts in the specification phase [96]. These questions enables assessing the correctness and completeness of the representation – **correctness** refers to whether the definitions of concepts and relationships accurately reflect the real-world domain (in this context, metabolism), while **completeness** concerns whether all necessary concepts and relationships have been captured to fully represent the domain knowledge.

The evaluation phase should also assess the overall quality of the generated model, which can be

done by comparing the application output with manually curated models from well-studied organisms, such as *Escherichia coli*. This evaluation approach is commonly adopted in the literature [28, 78–80], as curated models are the most complete available reconstructions of an organism and are supported by scientific publications that offer valuable information for evaluation. In our scenario, the pipeline will involve: (1) generating a draft model from the genome of the selected organism; (2) enhancing the model using the ontology-based application; and (3) comparing the resulting model with a curated reference. To ensure a more comprehensive assessment of the application's effectiveness, the comparison should include models of varying quality (both highly and poorly curated) from diverse organisms (e.g., eukaryotes and prokaryotes, well-studied and less-studied).

Although the comparison criteria for this project are not yet fully defined, they should probably include the number and presence or absence of metabolites, genes, and reactions [10, 79], the ability to directly perform FBA [28, 80], and reports from tools such as FROG [65] and MEMOTE [43], which provide additional insights into the metabolic network (e.g., mass and charge balancing, stoichiometric consistency, FVA, and gene/reaction deletion fluxes).

The number of genes, metabolites, and reactions correctly mapped and unified can be compared with other tools, such as MetaNetX and mergem, especially in template-based reconstructions. Additionally, analysis by domain experts is crucial to provide qualitative insights into model explainability, report quality, application usability, and the integration of the solution within the GEM reconstruction workflow.

3.5 Challenges and Limitations

As previously described, the ontology-based solution proposed in this project assumes that the input will be a draft model – an SBML file generated by another tool – containing the metabolic network to be enriched and refined. Consequently, the quality of the output will depend on the completeness and accuracy of the information available in the draft model. The input must include sufficient details to allow the application to retrieve relevant information from biochemical databases. Throughout the project, mechanisms will be developed to gather as much additional information as possible to enrich the model; however, some cases may remain unresolved due to time and resource constraints.

For example, during manual curation, genes and reactions may be included in the model based on evidence from the scientific literature. While part of this information is structured and available in databases like PubChem [53] and FORUM [7], much of it remains accessible only through the scientific publications. Although current Natural Language Processing (NLP) techniques [102, 103] and Large Language Models (LLMs) [104, 105] have demonstrated potential in processing such content and extracting relevant knowledge, this task is complex and would require a dedicated effort to be properly addressed.

Although many public biochemical databases are available, automatically retrieving data from them depends on several factors, including access permissions, licensing terms, usage limits, API endpoints, and online availability. These constraints can directly impact the application's performance; therefore, database selection must carefully consider these limitations.

Another key challenge to be addressed in this project is the development of a user-friendly application. As previously mentioned, many existing tools have not gained widespread adoption due to poor usability and insufficient documentation [5, 24, 83], which increases the learning curve and

discourages domain experts from using them. While current time and resource constraints limit the development of a fully featured software solution, the proof-of-concept implementation – including architectural decisions – should prioritize user experience and support future extensibility.

Chapter 4

Project Timeline and Work Plan

The previous chapters described the project proposal that seeks to streamline the GEM reconstruction process and contribute to the development of high-quality GEMs. This chapter presents a roadmap outlining the main activities to be executed in this project, as summarized in Figure 4.1.

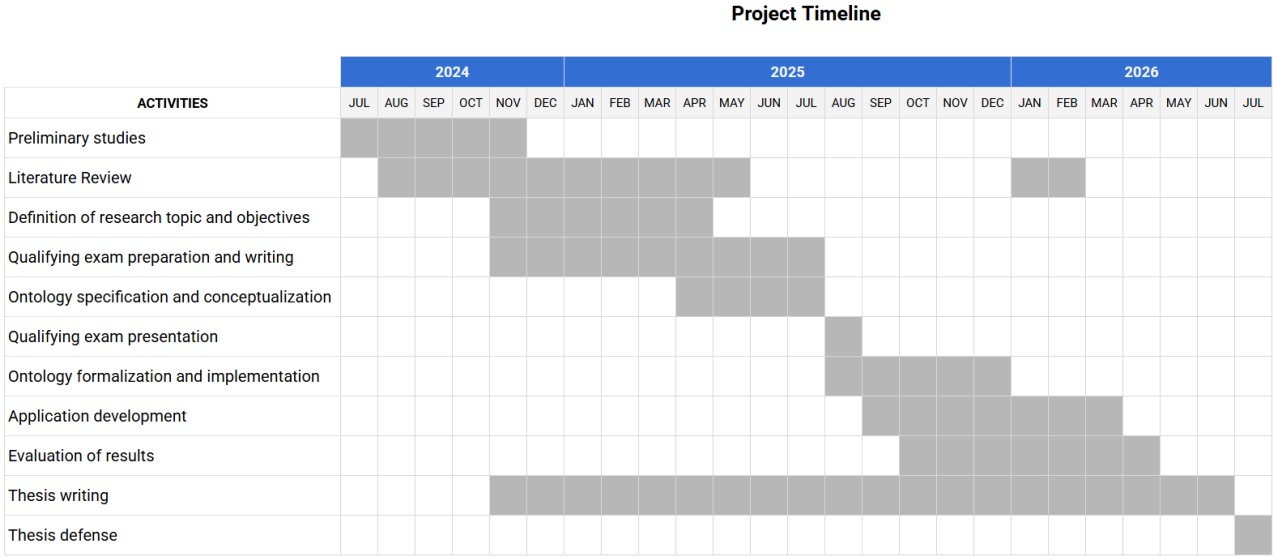


Figure 4.1: Overview of main activities and timeline.

4.1 Activities Description

The key activities planned for the development of this project include:

- **Preliminary Studies** – Preliminary studies and initial discussions with domain experts focused on understanding metabolism, the GEM reconstruction process, and the databases and tools commonly used to support this workflow. These efforts provided a clearer view of the main challenges faced by domain experts and underscored the need for a solution capable of effectively addressing them.
- **Literature Review** – The literature review provided a comprehensive knowledge about the domain and existing solutions (databases, tools and ontologies). The most intensive phase of the review was conducted during the past year. A shorter, second phase will be carried out in the

coming year to ensure an updated overview of recent developments before publishing the results.

- **Definition of research topic and objectives** – Following the preliminary studies, a series of discussions with domain experts was conducted to define a specific research topic and establish the project objectives. During this phase, an informal article outlining the GEM reconstruction process and its key challenges was written to align the domain understanding between the team and the experts. Competency questions were also formulated to clarify the knowledge that the ontology should represent.

- **Qualifying exam preparation and writing** – During this stage, the project proposal (i.e., this document) was formalized to detail the development plan and the strategies to evaluate the project's outcomes.

- **Ontology specification and conceptualization** – Once the project objectives and scope were defined, the ontology specification began with the extraction of concepts and relationships from the competency questions. These elements were then informally described, and a conceptual diagram (Figure 3.1) was created to support discussions with domain experts. Their feedback was crucial for refining the model and shaping the initial version of the ontology.

- **Qualifying exam presentation** – The presentation of this research project to the examination committee.

- **Ontology formalization and implementation** – This phase involves the formalization of the concepts and relationships in OWL, using Protégé as the development environment. Having the ontology in a machine-readable format, it should be populated with metabolic model data and augmented with information retrieved from biochemical databases.

- **Application development** – Application development to enable the interaction between the user (domain expert) and the ontology. This stage should occur in parallel with the ontology formalization and evaluation, since the application will facilitate the identification of problems in the ontology.

- **Evaluation of results** – Definition of assessment criteria involving application output analysis and domain expert feedback. The evaluation results should be used to refine the ontology and improve the application, while also providing an overall assessment of project outcomes.

- **Research writing and documentation** – Documentation of research activities and project outcomes. It must include recording ontology design decisions, the application development process, and evaluation results. The documentation will serve as a basis for scientific publications, and will also be incorporated into the final thesis.

- **Thesis defense** – Thesis defense before the examination committee, planned for July 2026.

A short research visit (one month) to the University of Lyon at the end of March 2025 facilitated communication with domain experts and accelerated decision-making processes, particularly in defining the project objectives and initiating the domain conceptualization. A second visit is planned for the second semester of this year, during which the first results of the project are expected to be analyzed.

Bibliography

1. Judge, A. & Dodd, M. S. Metabolism. *Essays in Biochemistry* **64,** 607–647. ISSN: 0071-1365. eprint: https://portlandpress.com/essaysbiochem/article-pdf/64/4/607/894527/ebc-2019-0041c.pdf. https://doi.org/10.1042/EBC20190041 (Aug. 2020).
2. Canuto, G. A. B. *et al.* Metabolômica: definições, estado-da-arte e aplicações representativas. *Química Nova* **41,** 75–91. ISSN: 0100-4042. eprint: http://www.scielo.br/pdf/qn/v41n1/0100-4042-qn-41-01-0075.pdf (2018).
3. Ferrarini, M. G. *et al.* Insights on the virulence of swine respiratory tract mycoplasmas through genome-scale metabolic modeling. *BMC Genomics* **17,** 353. ISSN: 1471-2164. https://doi.org/10.1186/s12864-016-2644-z (2016).
4. Julien-Laferriere, A. *Models and algorithms applied to metabolism: from revealing the responses to perturbations towards the design of microbial consortia* Theses (Université de Lyon, Dec. 2016). https://theses.hal.science/tel-01394113.
5. Wang, L., Dash, S., Ng, C. Y. & Maranas, C. D. A review of computational tools for design and reconstruction of metabolic pathways. *Synthetic and Systems Biotechnology* **2,** 243–252. ISSN: 2405-805X. https://www.sciencedirect.com/science/article/pii/S2405805X17300820 (2017).
6. Bernard, T. *et al.* Reconciliation of metabolites and biochemical reactions for metabolic networks. *Briefings in Bioinformatics* **15,** 123–135. ISSN: 1467-5463. eprint: https://academic.oup.com/bib/article-pdf/15/1/123/456230/bbs058.pdf. https://doi.org/10.1093/bib/bbs058 (Nov. 2012).
7. Delmas, M. *et al.* FORUM: building a Knowledge Graph from public databases and scientific literature to extract associations between chemicals and diseases. *Bioinformatics* **37,** 3896–3904. ISSN: 1367-4803. eprint: https://academic.oup.com/bioinformatics/article-pdf/37/21/3896/50337302/btab627.pdf. https://doi.org/10.1093/bioinformatics/btab627 (Sept. 2021).
8. Hoyt, C. T. *et al.* Unifying the Identification of Biomedical Entities with the Bioregistry. *bioRxiv*. eprint: https://www.biorxiv.org/content/early/2022/10/22/2022.07.08.499378.full.pdf. https://www.biorxiv.org/content/early/2022/10/22/2022.07.08.499378 (2022).
9. Hari, A., Zarrabi, A. & Lobo, D. mergem: merging, comparing, and translating genome-scale metabolic models using universal identifiers. *NAR Genomics and Bioinformatics* **6,** 1–21. ISSN: 2631-9268. eprint: https://academic.oup.com/nargab/article-pdf/6/1/lqae010/56553896/lqae010.pdf. https://doi.org/10.1093/nargab/lqae010 (Feb. 2024).
10. Hsieh, Y. E., Tandon, K., Verbruggen, H. & Nikoloski, Z. Comparative analysis of metabolic models of microbial communities reconstructed from automated tools and consensus approaches. *npj Systems Biology and Applications* **10,** 54. ISSN: 2056-7189. https://doi.org/10.1038/s41540-024-00384-y (2024).

11. Pham, N. *et al.* Consistency, Inconsistency, and Ambiguity of Metabolite Names in Biochemical Databases Used for Genome-Scale Metabolic Modelling. *Metabolites* **9.** ISSN: 2218-1989. https://www.mdpi.com/2218-1989/9/2/28 (2019).

12. Pfau, T., Pacheco, M. P. & Sauter, T. Towards improved genome-scale metabolic network reconstructions: unification, transcript specificity and beyond. *Briefings in Bioinformatics* **17,** 1060–1069. ISSN: 1467-5463. eprint: https://academic.oup.com/bib/article-pdf/17/6/1060/8280351/bbv100.pdf. https://doi.org/10.1093/bib/bbv100 (Nov. 2015).

13. Pellison, F. C. *et al.* Data Integration in the Brazilian Public Health System for Tuberculosis: Use of the Semantic Web to Establish Interoperability. *JMIR Med Inform* **8,** e17176. ISSN: 2291-9694. http://www.ncbi.nlm.nih.gov/pubmed/32628611 (July 2020).

14. Pereira, D. L. N. C., Pessoa, A., Salvador, L. & Wassermann, R. Integração semântica das bases de dados do Sistema Único de Saude: um estudo de caso com o município de São Paulo. *CEUR Workshop Proceedings* **2728,** 61–74. ISSN: 1614-0073. eprint: https://repositorio.usp.br/directbitstream/d11a43cb-3c13-46a0-ad36-64a565b33410/3013878.pdf. https://repositorio.usp.br/item/003013878 (2020).

15. Emygdio, J. L. *Interoperabilidade semântica orientada por ontologia para a Ciência da Informação: a metodologia Onto4All-Interoperability como resultado de estudo de caso no domínio de energia* PhD thesis (Escola de Ciência da Informação - Universidade Federal de Minas Gerais, 2021). http://hdl.handle.net/1843/42050.

16. Bernasconi, A., Guizzardi, G., Pastor, O. & Storey, V. C. Semantic interoperability: ontological unpacking of a viral conceptual model. *BMC Bioinformatics* **23,** 491. https://doi.org/10.1186/s12859-022-05022-0 (Nov. 2022).

17. Guizzardi, G. Ontology, Ontologies and the "I" of FAIR. *Data Intelligence* **2,** 181–191. ISSN: 2641-435X. eprint: https://direct.mit.edu/dint/article-pdf/2/1-2/181/1893393/dint_a_00040.pdf. https://doi.org/10.1162/dint_a_00040 (Jan. 2020).

18. Ashburner, M. *et al.* Gene Ontology: tool for the unification of biology. *Nature Genetics* **25,** 25–29. ISSN: 1546-1718. https://doi.org/10.1038/75556 (2000).

19. Degtyarenko, K. *et al.* ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36,** D344–D350. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/36/suppl_1/D344/7627418/gkm791.pdf. https://doi.org/10.1093/nar/gkm791 (Oct. 2007).

20. Courtot, M. *et al.* Controlled vocabularies and semantics in systems biology. *Molecular Systems Biology* **7,** 543. eprint: https://www.embopress.org/doi/pdf/10.1038/msb.2011.77. https://www.embopress.org/doi/abs/10.1038/msb.2011.77 (2011).

21. Altaf-Ul-Amin, M., Kanaya, S. & Mohamed-Hussein, Z.-A. in *Encyclopedia of Bioinformatics and Computational Biology* (eds Ranganathan, S., Gribskov, M., Nakai, K. & Schönbach, C.) 489–503 (Academic Press, Oxford, 2019). ISBN: 978-0-12-811432-2. https://www.sciencedirect.com/science/article/pii/B9780128096338201404.

22. Fang, X., Lloyd, C. & Palsson, B. Reconstructing organisms in silico: genome-scale models and their emerging applications. *Nature Reviews Microbiology* **18,** 731–743 (Sept. 2020).

23. Amara, A. *et al.* Networks and Graphs Discovery in Metabolomics Data Analysis and Interpretation. *Frontiers in Molecular Biosciences* **9.** ISSN: 2296-889X. https://www.frontiersin.org/journals/molecular-biosciences/articles/10.3389/fmolb.2022.841373 (2022).

24. Ankrah, N. Y. D. *et al.* Enhancing Microbiome Research through Genome-Scale Metabolic Modeling. *mSystems* **6,** e00599–21. eprint: https://journals.asm.org/doi/pdf/10.1128/msystems.00599-21. https://journals.asm.org/doi/abs/10.1128/msystems.00599-21 (2021).

25. Passi, A. *et al.* Genome-Scale Metabolic Modeling Enables In-Depth Understanding of Big Data. *Metabolites* **12.** ISSN: 2218-1989. https://www.mdpi.com/2218-1989/12/1/14 (2022).

26. Thiele, I. & Palsson, B. Ø. A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols* **5,** 93–121. ISSN: 1750-2799. https://doi.org/10.1038/nprot.2009.203 (2010).

27. Machado, D., Andrejev, S., Tramontano, M. & Patil, K. R. Fast automated reconstruction of genome-scale metabolic models for microbial species and communities. *Nucleic Acids Research* **46,** 7542–7553. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/46/15/7542/25689981/gky537.pdf. https://doi.org/10.1093/nar/gky537 (June 2018).

28. Seaver, S. M. D. *et al.* The ModelSEED Biochemistry Database for the integration of metabolic annotations and the reconstruction, comparison and analysis of metabolic models for plants, fungi and microbes. *Nucleic Acids Research* **49,** D575–D588. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/49/D1/D575/35364065/gkaa746.pdf. https://doi.org/10.1093/nar/gkaa746 (Sept. 2020).

29. Ferreira, G. C. in *Encyclopedia of Biological Chemistry III (Third Edition)* (ed Jez, J.) Third Edition, 356–362 (Elsevier, Oxford, 2021). ISBN: 978-0-12-822040-5. https://www.sciencedirect.com/science/article/pii/B9780128096338212673.

30. Fang, G., Bhardwaj, N., Robilotto, R. & Gerstein, M. B. Getting Started in Gene Orthology and Functional Analysis. *PLOS Computational Biology* **6,** 1–8. https://doi.org/10.1371/journal.pcbi.1000703 (Mar. 2010).

31. Wang, H. *et al.* RAVEN 2.0: A versatile toolbox for metabolic network reconstruction and a case study on Streptomyces coelicolor. *PLOS Computational Biology* **14,** 1–17. https://doi.org/10.1371/journal.pcbi.1006541 (Oct. 2018).

32. Kanehisa, M. & Goto, S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res* **28,** 27–30. http://www.ncbi.nlm.nih.gov/pubmed/10592173 (Jan. 2000).

33. Chang, A. *et al.* BRENDA, the ELIXIR core data resource in 2021: new developments and updates. *Nucleic Acids Research* **49,** D498–D508. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/49/D1/D498/35364686/gkaa1025.pdf. https://doi.org/10.1093/nar/gkaa1025 (Nov. 2020).

34. King, Z. A. *et al.* BiGG Models: A platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Research* **44,** D515–D522. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/44/D1/D515/16661243/gkv1049.pdf. https://doi.org/10.1093/nar/gkv1049 (Oct. 2015).

35. Bernstein, D. B., Sulheim, S., Almaas, E. & Segrè, D. Addressing uncertainty in genome-scale metabolic model reconstruction and analysis. *Genome Biology* **22,** 64. ISSN: 1474-760X. https://doi.org/10.1186/s13059-021-02289-z (2021).

36. Moretti, S., Tran, V. D. T., Mehl, F., Ibberson, M. & Pagni, M. MetaNetX/MNXref: unified namespace for metabolites and biochemical reactions in the context of metabolic models. *Nucleic Acids Research* **49,** D570–D574. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/49/D1/D570/35364710/gkaa992.pdf. https://doi.org/10.1093/nar/gkaa992 (Nov. 2020).

37. Keating, S. M. *et al.* SBML Level 3: an extensible format for the exchange and reuse of biological models. *Molecular Systems Biology* **16,** e9110. eprint: https://www.embopress.org/doi/pdf/10.15252/msb.20199110. https://www.embopress.org/doi/abs/10.15252/msb.20199110 (2020).

38. Hastings, J. *et al.* ChEBI in 2016: Improved services and an expanding collection of metabolites. *Nucleic acids research* **44,** D1214–9. ISSN: 0305-1048. https://europepmc.org/articles/PMC4702775 (Jan. 2016).

39. Ebrahim, A., Lerman, J. A., Palsson, B. O. & Hyduke, D. R. COBRApy: COnstraints-Based Reconstruction and Analysis for Python. *BMC Systems Biology* **7,** 74+. ISSN: 1752-0509. http://dx.doi.org/10.1186/1752-0509-7-74 (Aug. 2013).

40. Heirendt, L. *et al.* Creation and analysis of biochemical constraint-based models using the COBRA Toolbox v.3.0. *Nature Protocols* **14,** 639–702. ISSN: 17502799. https://doi.org/10.1038/s41596-018-0098-2 (2019).

41. Orth, J., Thiele, I. & Palsson, B. What is ux balance analysis? *Nature biotechnology* **28,** 245–8 (Mar. 2010).

42. Palsson, B. Ø. in *Systems Biology: Constraint-based Reconstruction and Analysis* 151–171 (Cambridge University Press, 2015).

43. Lieven, C. *et al.* MEMOTE for standardized genome-scale metabolic model testing. *Nature Biotechnology* **38** (2020).

44. Shiratsubaki, I. S. *et al.* Genome-scale metabolic models highlight stage-specific differences in essential metabolic pathways in Trypanosoma cruzi. *PLOS Neglected Tropical Diseases* **14,** 1–20. https://doi.org/10.1371/journal.pntd.0008728 (Oct. 2020).

45. Caspi, R. *et al.* The MetaCyc database of metabolic pathways and enzymes - a 2019 update. *Nucleic Acids Research* **48,** D445–D453. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/48/D1/D445/31697668/gkz862.pdf. https://doi.org/10.1093/nar/gkz862 (Oct. 2019).

46. Bansal, P. *et al.* Rhea, the reaction knowledgebase in 2022. *Nucleic Acids Research* **50,** D693–D700. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/50/D1/D693/42058388/gkab1016.pdf. https://doi.org/10.1093/nar/gkab1016 (Nov. 2021).

47. Milacic, M. *et al.* The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Research* **52,** D672–D678. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/52/D1/D672/55039828/gkad1025.pdf. https://doi.org/10.1093/nar/gkad1025 (Nov. 2023).

48. Haug, K. *et al.* MetaboLights—an open-access general-purpose repository for metabolomics studies and associated meta-data. *Nucleic Acids Research* **41,** D781–D786. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/41/D1/D781/18785371/gks1004.pdf. https://doi.org/10.1093/nar/gks1004 (Oct. 2012).

49. Conroy, M. J. *et al.* LIPID MAPS: update to databases and tools for the lipidomics community. *Nucleic Acids Research* **52,** D1677–D1682. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/52/D1/D1677/55040242/gkad896.pdf. https://doi.org/10.1093/nar/gkad896 (Oct. 2023).

50. Aimo, L. *et al.* The SwissLipids knowledgebase for lipid biology. *Bioinformatics* **31,** 2860–2866. ISSN: 1367-4803. eprint: https://academic.oup.com/bioinformatics/article-pdf/31/17/2860/49035148/bioinformatics_31_17_2860.pdf. https://doi.org/10.1093/bioinformatics/btv285 (May 2015).

51. Wishart, D. S. *et al.* HMDB 5.0: the Human Metabolome Database for 2022. *Nucleic Acids Research* **50,** D622–D631. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/50/D1/D622/42058215/gkab1062.pdf. https://doi.org/10.1093/nar/gkab1062 (Nov. 2021).

52. Consortium, T. U. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research* **51,** D523–D531. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/51/D1/D523/48441158/gkac1052.pdf. https://doi.org/10.1093/nar/gkac1052 (Nov. 2022).

53. Kim, S. *et al.* PubChem 2023 update. *Nucleic Acids Research* **51,** D1373–D1380. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/51/D1/D1373/48441598/gkac956.pdf. https://doi.org/10.1093/nar/gkac956 (Oct. 2022).

54. Agrawal, A. *et al.* WikiPathways 2024: next generation pathway database. *Nucleic Acids Research* **52,** D679–D689. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/52/D1/D679/55040703/gkad960.pdf. https://doi.org/10.1093/nar/gkad960 (Nov. 2023).

55. Fu, G. *et al.* PubChemRDF: towards the semantic annotation of PubChem compound and substance databases. *Journal of Cheminformatics* **7,** 34. ISSN: 1758-2946. https://doi.org/10.1186/s13321-015-0084-4 (2015).

56. Shanmugasundram, A. *et al.* TriTrypDB: An integrated functional genomics resource for kinetoplastida. *PLOS Neglected Tropical Diseases* **17,** 1–20. https://doi.org/10.1371/journal.pntd.0011058 (Jan. 2023).

57. Bornstein, B. J., Keating, S. M., Jouraku, A. & Hucka, M. LibSBML: An API Library for SBML. *Bioinformatics* **24,** 880–881. ISSN: 1367-4803. http://dx.doi.org/10.1093/bioinformatics/btn051 (2008).

58. Thiele, S., von Kamp, A., Bekiaris, P. S., Schneider, P. & Klamt, S. CNApy: a CellNetAnalyzer GUI in Python for analyzing and designing metabolic networks. *Bioinformatics* **38,** 1467–1469. ISSN: 1367-4803. eprint: https://academic.oup.com/bioinformatics/article-pdf/38/5/1467/49008942/btab828.pdf. https://doi.org/10.1093/bioinformatics/btab828 (Dec. 2021).

59. Beber, M. E. *et al.* eQuilibrator 3.0: a database solution for thermodynamic constant estimation. *Nucleic Acids Research* **50,** D603–D609. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/50/D1/D603/42057662/gkab1106.pdf. https://doi.org/10.1093/nar/gkab1106 (Nov. 2021).

60. Henry, C. S. *et al.* High-throughput generation, optimization and analysis of genome-scale metabolic models. *Nature Biotechnology* **28,** 977–982. ISSN: 1546-1696. https://doi.org/10.1038/nbt.1672 (2010).

61. Arkin, A. P. *et al.* KBase: The United States Department of Energy Systems Biology Knowledgebase. *Nature Biotechnology* **36** (July 2018).

62. Cottret, L. *et al.* MetExplore: collaborative edition and exploration of metabolic networks. *Nucleic Acids Research* **46,** W495–W502. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/46/W1/W495/25110581/gky301.pdf. https://doi.org/10.1093/nar/gky301 (Apr. 2018).

63. Karp, P. D. *et al. Pathway Tools version 28.0: Integrated Software for Pathway/Genome Informatics and Systems Biology* 2024. arXiv: 1510.03964 [q-bio.GN]. https://arxiv.org/abs/1510.03964.

64. Capela, J. *et al.* merlin, an improved framework for the reconstruction of high-quality genome-scale metabolic models. *Nucleic Acids Research* **50,** 6052–6066. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/50/11/6052/49800920/gkac459.pdf. https://doi.org/10.1093/nar/gkac459 (June 2022).

65. Raman, K. *et al.* FROG Analysis Ensures the Reproducibility of Genome Scale Metabolic Models. *bioRxiv.* eprint: https://www.biorxiv.org/content/early/2024/09/26/2024.09.24.614797.full.pdf. https://www.biorxiv.org/content/early/2024/09/26/2024.09.24.614797 (2024).

66. King, Z. A. *et al.* Escher: A Web Application for Building, Sharing, and Embedding Data-Rich Visualizations of Biological Pathways. *PLOS Computational Biology* **11,** 1–13. https://doi.org/10.1371/journal.pcbi.1004321 (Aug. 2015).

67. Hari, A. & Lobo, D. Fluxer: a web application to compute, analyze and visualize genome-scale metabolic flux networks. *Nucleic Acids Research* **48,** W427–W435. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/48/W1/W427/33433441/gkaa409.pdf. https://doi.org/10.1093/nar/gkaa409 (May 2020).

68. Duvaud, S. *et al.* Expasy, the Swiss Bioinformatics Resource Portal, as designed by its users. *Nucleic Acids Research* **49,** W216–W227. ISSN: 0305-1048. eprint: https://academic.oup.com/nar/article-pdf/49/W1/W216/38842229/gkab225.pdf. https://doi.org/10.1093/nar/gkab225 (Apr. 2021).

69. Weininger, D. SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences* **28,** 31–36. eprint: https://doi.org/10.1021/ci00057a005. https://doi.org/10.1021/ci00057a005 (1988).

70. McNaught, A. The IUPAC international chemical identifier: InChl-A new standard for molecular informatics. *Chemistry international* **28,** 12–14. https://api.semanticscholar.org/CorpusID:89057853 (2006).

71. Dalby, A. *et al.* Description of several chemical structure file formats used by computer programs developed at Molecular Design Limited. *Journal of Chemical Information and Computer Sciences* **32,** 244–255. https://doi.org/10.1021/ci00007a012 (1992).

72. Hattori, M., Okuno, Y., Goto, S. & Kanehisa, M. Development of a Chemical Structure Comparison Method for Integrated Analysis of Chemical and Genomic Information in the Metabolic Pathways. *Journal of the American Chemical Society* **125,** 11853–11865. https://doi.org/10.1021/ja036030u (2003).

73. Kotera, M. *et al.* KCF-S: KEGG Chemical Function and Substructure for improved interpretability and prediction in chemical bioinformatics. *BMC Systems Biology* **7,** S2. https://doi.org/10.1186/1752-0509-7-S6-S2 (2013).

74. Cottret, L. *et al. Enumerating Precursor Sets of Target Metabolites in a Metabolic Network* in *Algorithms in Bioinformatics* (eds Crandall, K. A. & Lagergren, J.) (Springer Berlin Heidelberg, Berlin, Heidelberg, 2008), 233–244. ISBN: 978-3-540-87361-7.

75. Lacroix, V., Cottret, L., Thébault, P. & Sagot, M.-F. An Introduction to Metabolic Networks and Their Structural Analysis. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* **5,** 594–617 (2008).

76. Kumar, A., Suthers, P. F. & Maranas, C. D. MetRxn: a knowledgebase of metabolites and reactions spanning metabolic models and databases. *BMC Bioinformatics* **13,** 6. https://doi.org/10.1186/1471-2105-13-6 (2012).

77. Strömbäck, L. & Lambrix, P. Representations of molecular pathways: an evaluation of SBML, PSI MI and BioPAX. *Bioinformatics* **21,** 4401–4407. ISSN: 1367-4803. eprint: https://academic.oup.com/bioinformatics/article-pdf/21/24/4401/50566203/bioinformatics_21_24_4401.pdf. https://doi.org/10.1093/bioinformatics/bti718 (Oct. 2005).

78. Di Filippo, M., Damiani, C. & Pescini, D. GPRuler: Metabolic gene-protein-reaction rules automatic reconstruction. *PLOS Computational Biology* **17,** 1–25. https://doi.org/10.1371/journal.pcbi.1009550 (Nov. 2021).

79. Mendoza, S. N., Olivier, B. G., Molenaar, D. & Teusink, B. A systematic assessment of current genome-scale metabolic reconstruction tools. *Genome Biology* **20,** 158. ISSN: 1474-760X. https://doi.org/10.1186/s13059-019-1769-1 (2019).

80. Zimmermann, J., Kaleta, C. & Waschina, S. gapseq: informed prediction of bacterial metabolic pathways and reconstruction of accurate metabolic models. *Genome Biology* **22,** 81. ISSN: 1474-760X. https://doi.org/10.1186/s13059-021-02295-1 (2021).

81. Kim, S. *et al.* Literature information in PubChem: associations between PubChem records and scientific articles. *Journal of Cheminformatics* **8,** 32. ISSN: 1758-2946. https://doi.org/10.1186/s13321-016-0142-6 (2016).

82. Carey, M. A., Dräger, A., Beber, M. E., Papin, J. A. & Yurkovich, J. T. Community standards to facilitate development and address challenges in metabolic modeling. *Molecular Systems Biology* **16,** e9235. eprint: https://www.embopress.org/doi/pdf/10.15252/msb.20199235. https://www.embopress.org/doi/abs/10.15252/msb.20199235 (2020).

83. Beust, C., Becker, E., Théret, N. & Dameron, O. BioPAX in 2024: Where we are and where we are heading. *Computational and Structural Biotechnology Journal* **23,** 3999–4010. ISSN: 2001-0370. https://www.sciencedirect.com/science/article/pii/S2001037024003660 (2024).

84. Smith, B. in *The Blackwell guide to the philosophy of computing and information* (ed Floridi, L.) 155–166 (Blackwell, 2003).

85. Guizzardi, G. & Guarino, N. Explanation, semantics, and ontology. *Data and Knowledge Engineering* **153,** 102325. ISSN: 0169-023X. https://www.sciencedirect.com/science/article/pii/S0169023X24000491 (2024).

86. Strömert, P., Hunold, J., Castro, A., Neumann, S. & Koepler, O. *Pure and Applied Chemistry* **94,** 605–622. https://doi.org/10.1515/pac-2021-2007 (2022).

87. Hitzler, P., Krötzsch, M., Parsia, B., Patel-Schneider, P. F. & Rudolph, S. *OWL 2 Web Ontology Language: Primer (Second Edition)* W3C Recommendation. Latest version available at http://www.w3.org/TR/owl2-primer/. Dec. 2012. http://www.w3.org/TR/2012/REC-owl2-primer-20121211/.

88. Cyganiak, R., Wood, D. & Lanthaler, M. *RDF 1.1 Concepts and Abstract Syntax* W3C Recommendation. Latest published version available at http://www.w3.org/TR/rdf11-concepts/. Feb. 2014. http://www.w3.org/TR/2014/REC-rdf11-concepts-20140225/.

89. Guizzardi, G., Pastor, O. & Storey, V. C. Thinking Fast and Slow in Software Engineering. *IEEE Software* **40,** 139–142. ISSN: 1937-4194. https://doi.ieeecomputersociety.org/10.1109/MS.2023.3306132 (Nov. 2023).

90. Neuhaus, F. & Hastings, J. Ontology development is consensus creation, not (merely) representation. *Applied Ontology* **17,** 495–513. eprint: https://journals.sagepub.com/doi/pdf/10.3233/AO-220273. https://journals.sagepub.com/doi/abs/10.3233/AO-220273 (2022).

91. Fernández-López, M., Gómez-Pérez, A. & Juristo, N. *METHONTOLOGY: From Ontological Art Towards Ontological Engineering* in *AAAI Conference on Artificial Intelligence* (1997). https://api.semanticscholar.org/CorpusID:10550105.

92. Noy, N. & McGuinness, D. Ontology Development 101: A Guide to Creating Your First Ontology. *Knowledge Systems Laboratory* **32** (Jan. 2001).

93. Suárez-Figueroa, M. C., Gómez-Pérez, A. & Fernández-López, M. in *Ontology Engineering in a Networked World* (eds Suárez-Figueroa, M. C., Gómez-Pérez, A., Motta, E. & Gangemi, A.) 9–34 (Springer Berlin Heidelberg, Berlin, Heidelberg, 2012). ISBN: 978-3-642-24794-1. https://doi.org/10.1007/978-3-642-24794-1_2.

94. Falbo, R. d. A. SABiO: Systematic approach for building ontologies. *CEUR Workshop Proceedings* **1301** (Jan. 2014).

95. Mendonça, F. M. & Almeida, M. B. OntoForInfoScience: a detailed methodology for construction of ontologies and its application in the blood domain. *Brazilian Journal of Information Science: research trends* **10.** https://revistas.marilia.unesp.br/index.php/bjis/article/view/5426 (Mar. 2016).

96. Monfardini, G. K. Q., Salamon, J. S. & Barcellos, M. P. *Use of Competency Questions in Ontology Engineering: A Survey* in *Conceptual Modeling* (eds Almeida, J. P. A., Borbinha, J., Guizzardi, G., Link, S. & Zdravkovic, J.) (Springer Nature Switzerland, Cham, 2023), 45–64. ISBN: 978-3-031-47262-6.

97. Otte, J. N., Beverley, J. & Ruttenberg, A. BFO: Basic Formal Ontology. *Applied ontology* **17,** 17–43 (2022).

98. Hucka, M. *et al. Journal of Integrative Bioinformatics* **16,** 20190021. https://doi.org/10.1515/jib-2019-0021 (2019).

99. Wilkinson, M. D. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* **3,** 160018. https://doi.org/10.1038/sdata.2016.18 (Mar. 2016).

100. Pereira, D. L. N. C. *Integração semântica das bases de dados do Sistema Único de Saude: um estudo de caso com o município de São Paulo* MA thesis (Instituto de Matemática e Estatística, Universidade de São Paulo, 2019). https://www.teses.usp.br/teses/disponiveis/45/45134/tde-19032020-134428/pt-br.php.

101. García S., A. *et al.* Assessing the value of ontologically unpacking a conceptual model for human genomics. *Information Systems* **118,** 102242. ISSN: 0306-4379. https://www.sciencedirect.com/science/article/pii/S0306437923000789 (2023).

102. Santana, B. *et al.* A survey on narrative extraction from textual data. *Artificial Intelligence Review* **56,** 8393–8435. ISSN: 1573-7462. https://doi.org/10.1007/s10462-022-10338-7 (2023).

103. Mann, V., Viswanath, S., Vaidyaraman, S., Balakrishnan, J. & Venkatasubramanian, V. SUSIE: Pharmaceutical CMC ontology-based information extraction for drug development using machine learning. *Computers & Chemical Engineering* **179,** 108446. ISSN: 0098-1354. https://www.sciencedirect.com/science/article/pii/S0098135423003162 (2023).

104. Brown, T. *et al. Language Models are Few-Shot Learners* in *Advances in Neural Information Processing Systems* (eds Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M. & Lin, H.) **33** (Curran Associates, Inc., 2020), 1877–1901. https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

105. Buehler, M. J. Accelerating scientific discovery with generative knowledge extraction, graph-based representation, and multimodal intelligent graph reasoning. *Machine Learning: Science and Technology* **5,** 035083. https://dx.doi.org/10.1088/2632-2153/ad7228 (Sept. 2024).